



Pipeline

Scrape HackerNews (Show/Top/New), Reddit (r/artificial, r/MachineLearning, r/singularity, r/LocalLLaMA), TechCrunch AI, VentureBeat AI, MIT Tech Review, The Verge AI, Wired AI, ArXiv CS.AI

Filter 40+ AI keyword patterns — LLMs, alignment, safety, research, tooling

Dedup Title-similarity (45% threshold) merges cross-source variants

Score Authenticity = 0.5 base + 0.3 per extra source + diversity + HN score bonuses

Rank Sorted by authenticity. Below 0.25 threshold are dropped.

Authenticity Tiers

- VERIFIED 0.80 - 1.00** 3+ independent sources including media
- CONFIRMED 0.50 - 0.79** 2 sources or 1 high-quality media
- EMERGING 0.25 - 0.49** Single-source; passes AI keyword filter

Source Breakdown

Source	Articles	Category
ArXiv CS.AI	225	Media
TechCrunch AI	8	Media
HackerNews	7	Community
MIT Tech Review	3	Media
The Verge AI	2	Media

VERIFIED INTELLIGENCE — 1 story

VERI

Diversity is the Strength of the AI Crowd

ArXiv CS.AI

+1 source

80%

arXiv:2606.29661v1 Announce Type: new Abstract: Top AI forecasting systems are approaching superforecaster-level accuracy on future world events, but still rely primarily on off-the-shelf LLMs combined with forecasting-specific context gathering and scaffolding. We study how to...

<https://arxiv.org/abs/2606.29661>

CONFIRMED SIGNALS — 243 stories

CONF

Show HN: Appaca - AI Workspace for Operators

HackerNews

50%

HN 19

I'm a solo founder building AMA2, a messaging runtime made for AI agents. This is my first Show HN, so I'd really appreciate your feedback. What brought me this idea: At first, I was building an AI agent for solo creators that knows everything about you and can do business chore...

<https://www.appaca.ai/>

CONF

Show HN: Uselink - publish and collaborate on AI-generated HTML

HackerNews

50%

HN 8

No summary — see link for full article.

<https://uselink.app:443/>

CONF

Show HN: Xenoeye - analyze network without AI using netflow, PostgreSQL, Grafana

HackerNews

50%

HN 5

Sorry for the slightly truncated title. It should have been "Network traffic analysis and monitoring without AI, using netflow-family protocols, PostgreSQL or ClickHouse, Grafana, and some scripts". In 2026, it might seem a bit presumptuous to announce AI-free software on HN....

<https://github.com/vmxdev/xenoeye>

CONF

Show HN: Backtest a futures strategy described in plain English

HackerNews

50%

HN 5

No summary — see link for full article.

<https://askfutures.com>

CONF

Show HN: AI Music Generator Free and No Signup

HackerNews

50%

HN 4

No summary — see link for full article.

<https://music0.org/>

CONF

Show HN: Klorn-I built an email firewall because every AI inbox made mine louder

HackerNews

50%

HN 2

No summary — see link for full article.

<https://github.com/k08200/klorn>

CONF

Show HN: The UNESCO Tsunami Warning Emails Are Gone

HackerNews 50% HN 2

This key piece of tsunami warning and safety was discontinued this morning and evidently there's no way to get it back. :/
<https://lists.unesco.org/www/info/tsunami-information-ioc>

CONF

The AI jobs debate just got messier

TechCrunch AI 50%

A new report finds "high-intensity AI adopters" saw headcount increase 10.2%. Among those companies, entry-level headcount rose by 12%, countering the rhetoric that AI kills junior jobs.
<https://techcrunch.com/2026/06/29/the-ai-jobs-debate-just-got-messier/>

0

CONF

Vibe coding platform Base44 launches own model as AI startups seek defensibility

TechCrunch AI 50%

Wix-owned vibe coding platform Base44 has started rolling out its own AI model — with hopes that it will eventually outperform frontier models.
<https://techcrunch.com/2026/06/29/vibe-coding-platform-base44-launches-own-model-as-ai-...>

1

CONF

Gemini's personalized AI image generation is now free for US users

TechCrunch AI 50%

Google is expanding Gemini's personalized AI image generation to eligible free users in the U.S., allowing the chatbot to create images based on your interests and data from connected Google apps.
<https://techcrunch.com/2026/06/29/geminis-personalized-ai-image-generation-is-now-free-...>

2

CONF

Anthropic and Gov. Newsom forge deal allowing California government to use Claude at half price

TechCrunch AI 50%

As Anthropic forges a closer relationship with the state of California, the federal government has made an enemy out of the OpenAI rival.
<https://techcrunch.com/2026/06/29/anthropic-and-gov-newsom-forge-deal-allowing-californ...>

3

CONF

Arena, the AI leaderboard everyone uses, is now a \$100M business

TechCrunch AI 50%

The startup, which runs a popular free AI leaderboard, launched its commercial service just last September.
<https://techcrunch.com/2026/06/29/arena-the-ai-leaderboard-everyone-uses-is-now-a-100m-...>

4

CONF

Robot hand company settles Tesla trade secret suit and announces \$11M raise

TechCrunch AI 50%

The startup, Proception, is taking a unique approach to collecting training data to tackle one of the hardest problems in robotics: hands.
<https://techcrunch.com/2026/06/29/robot-hand-company-settles-tesla-trade-secret-suit-an...>

5

CONF

Omen AI's plan to optimize data centers is all wet

TechCrunch AI 50%

Omen AI raised a \$31 million Series A to monitor chip coolant and stop bacterial outbreaks in data centers.
<https://techcrunch.com/2026/06/29/omen-ais-plan-to-optimize-data-centers-is-all-wet/>

6 **CONF****Trump administration releases Anthropic Mythos to be used by more than 100 US companies, agencies****TechCrunch AI** 50%

Over 100 companies and government agencies are reportedly authorized to use Mythos 5, including their non-American employees.

<https://techcrunch.com/2026/06/26/trump-admin-releases-anthropic-mythos-to-be-used-by-m...>

7 **CONF****AI agents are not your “coworkers”****MIT Tech Review** 50%

This story originally appeared in The Algorithm, our weekly newsletter on AI. To get stories like this in your inbox first, sign up here. Imagine coming in to work to learn that a new underling will report to you. The worker is not a person but an AI tool—one that your company...

<https://www.technologyreview.com/2026/06/29/1139849/ai-agents-are-not-your-coworkers/>

8 **CONF****The Download: metric weaknesses and AI elephant warnings****MIT Tech Review** 50%

This is today's edition of The Download, our weekday newsletter that provides a daily dose of what's going on in the world of technology. The inevitable weakness of metrics There are plenty of useful things a metric can reveal. There are even more that it can obscure or corrupt....

<https://www.technologyreview.com/2026/06/29/1139834/the-download-metric-weaknesses-ai-e...>

9 **CONF****OpenAI is teasing new hardware... for Codex****The Verge AI** 50%

OpenAI is releasing some sort of device related to its AI-powered coding tool, Codex, on July 15th. In a video posted to X on Monday, OpenAI shows a square-shaped device with several buttons, alongside the caption, "Your favorite Codex shortcuts are getting an upgrade." This...

<https://www.theverge.com/ai-artificial-intelligence/959174/openai-codex-hardware-work-l...>

0 **CONF****Lawmakers want to ban AI companies from selling your health data****The Verge AI** 50%

A new proposal would ban the sale of Americans' health and location information to data brokers - including information people reveal to an AI chatbot like ChatGPT or Claude. In the coming weeks, Senator Elizabeth Warren (D-MA) and Representative Mary Gay Scanlon (D-PA) are...

<https://www.theverge.com/ai-artificial-intelligence/959033/health-location-data-protect...>

1 **CONF****GPTNT: Benchmarking Real-Time Collaboration Between Multimodal Agents on Keep Talking And Nobody Explodes****ArXiv CS.AI** 50%

arXiv:2606.28514v1 Announce Type: new Abstract: Multimodal models are increasingly deployed to solve tasks collaboratively with humans or other artificial agents. Existing benchmarks show that these models possess many of the required component capabilities, but the conditions...

<https://arxiv.org/abs/2606.28514>

2 CONF

Invariant Reasoning Directions in Latent Trajectories of Language Models

ArXiv CS.AI 50%

arXiv:2606.29164v1 Announce Type: cross Abstract: Latent reasoning models perform multi-step inference directly in hidden-state space, yet the structure of these latent reasoning trajectories remains poorly understood. We show that contrastive refinement signals between stronger...

<https://arxiv.org/abs/2606.29164>

3 CONF

Aristotelian Virtue Profiling of LLMs through Ethical Dilemmas

ArXiv CS.AI 50%

arXiv:2606.28683v1 Announce Type: new Abstract: Large Language Models (LLMs) often face ethical tradeoffs in which several responses may be defensible but express different priorities, such as fairness, honesty, courage, or restraint. We introduce VirtueMap, a framework for...

<https://arxiv.org/abs/2606.28683>

4 CONF

An AI agent for treatment reasoning over a biomedical tool universe

ArXiv CS.AI 50%

arXiv:2606.28692v1 Announce Type: new Abstract: Treatment reasoning underpins every therapeutic decision, integrating disease context, comorbidities, medications, contraindications, and evolving biomedical knowledge to select an appropriate therapy. It is inherently iterative:...

<https://arxiv.org/abs/2606.28692>

5 CONF

COMPASS: Grounding Composition-Intent Guidance in Unified Multimodal Models

ArXiv CS.AI 50%

arXiv:2606.28696v1 Announce Type: new Abstract: Composition is a high-level visual intent that governs where subjects are placed and how a scene is organized, yet current unified multimodal models remain unreliable at fine-grained composition recognition and struggle to turn...

<https://arxiv.org/abs/2606.28696>

6 CONF

BV-Blend: Uncertainty-Weighted Historical Baselines for Stable Critic-Free RL with Verifiable Rewards

ArXiv CS.AI 50%

arXiv:2606.28707v1 Announce Type: new Abstract: Critic-free reinforcement learning with verifiable rewards (RLVR), exemplified by Group Relative Policy Optimization (GRPO), avoids training a value function (critic) and reduces memory and compute overhead relative to critic-based...

<https://arxiv.org/abs/2606.28707>

7 CONF

Entity Binding Failures in Tool-Augmented Agents

ArXiv CS.AI 50%

arXiv:2606.30531v1 Announce Type: new Abstract: Tool-augmented language-model agents are often evaluated by whether they select the correct tool, produce valid API arguments, and complete the requested task. However, an agent may choose the right tool and still act on the wrong...

<https://arxiv.org/abs/2606.30531>

8 CONF

ComMem: Complementary Memory Systems for Test-Time Adaptation of Vision-Language Models

ArXiv CS.AI 50%

arXiv:2606.28719v1 Announce Type: new Abstract: Test-time adaptation (TTA) of vision-language models (VLMs) is essential for their robust deployment in dynamic, real-world environments. However, existing TTA methods often adapt locally without accumulating knowledge over time,...

<https://arxiv.org/abs/2606.28719>

9 CONF

Agent Safety Is Action Alignment

ArXiv CS.AI 50%

arXiv:2606.28739v1 Announce Type: new Abstract: Large language models increasingly act as agents: they call tools, move money, delete records, and send messages on a user's behalf. To keep them safe, practitioners imported the chatbot-era recipe (train the model to refuse unsafe...

<https://arxiv.org/abs/2606.28739>

0 CONF

Mechanistically Eliciting Latent Behaviors in Language Models

ArXiv CS.AI 50%

arXiv:2606.29604v1 Announce Type: cross Abstract: We aim to discover diverse, generalizable perturbations of LLM internals that can surface hidden behavioral modes. Such perturbations could help reshape model behavior and systematically evaluate potential risks. We introduce...

<https://arxiv.org/abs/2606.29604>

1 CONF

Diff-Based Code Corruption using LLMs for Large-Scale Bugfix Benchmarking

ArXiv CS.AI 50%

arXiv:2606.29088v1 Announce Type: cross Abstract: There are various benchmarks to evaluate bugfixing capabilities of Large Language Models. However, most widespread benchmarks do not fully reflect real-world bugfixing practices. They are small, weakening statistical reliability,...

<https://arxiv.org/abs/2606.29088>

2 CONF

Expert Evaluation of Clinical AI Tools on Real Point-of-Care Clinical Queries

ArXiv CS.AI 50%

arXiv:2606.28960v1 Announce Type: new Abstract: Physicians now pose millions of clinical questions to AI tools each week, yet these tools are evaluated largely on hypothetical or exam-style questions, not those actually asked in practice. We report a blinded evaluation built on...

<https://arxiv.org/abs/2606.28960>

3 CONF

Customized Generative AI Agent for Transportation Engineering Practice: A Development and Continued Pre-training Guideline

ArXiv CS.AI 50%

arXiv:2606.29014v1 Announce Type: new Abstract: Recent advancements in generative artificial intelligence (AI) and large language models (LLMs) have shown significant promise in automating complex reasoning, summarization, and question-answering tasks. However, the effectiveness...

<https://arxiv.org/abs/2606.29014>

4 **CONF****ACPO: Agent-Chained Policy Optimization for Multi-Agent Reinforcement Learning****ArXiv CS.AI 50%**

arXiv:2606.30072v1 Announce Type: new Abstract: Cooperative tasks in Multi-Agent Reinforcement Learning (MARL) require agents to collectively maximize a shared return. Under the Centralized Training with Decentralized Execution (CTDE) paradigm, policy gradients have remained...

<https://arxiv.org/abs/2606.30072>

5 **CONF****Towards Modality-Agnostic Medical Image Anomaly Detection: A Training-Free Manifold Refinement Approach****ArXiv CS.AI 50%**

arXiv:2604.19191v2 Announce Type: replace-cross Abstract: Deploying AI-based anomaly detection across diverse clinical imaging settings remains challenging because most existing methods rely on modality-specific architectures, anatomical priors, or extensive retraining, limiting...

<https://arxiv.org/abs/2604.19191>

6 **CONF****Managing the Human Fallback: Skill Investment Under Improving AI and Worker Mobility****ArXiv CS.AI 50%**

arXiv:2606.29111v1 Announce Type: new Abstract: When firms deploy autonomous AI, they must decide how much work to leave to the system and how much to keep workers engaged. This decision affects current output and future human capital. We develop a parsimonious two-period model...

<https://arxiv.org/abs/2606.29111>

7 **CONF****LoRA-Tuned Large Language Models for Dementia Detection via Multi-View Speech-Derived Features****ArXiv CS.AI 50%**

arXiv:2606.28445v1 Announce Type: cross Abstract: Early detection of dementia enables timely intervention, and reflecting cognitive impairment, spontaneous speech offers a non-invasive screening modality. Conventional approaches often focus on a single representational dimension...

<https://arxiv.org/abs/2606.28445>

8 **CONF****HiComm: Hierarchical Communication for Multi-agent Reinforcement Learning****ArXiv CS.AI 50%**

arXiv:2606.29126v1 Announce Type: new Abstract: Cooperative multi-agent reinforcement learning (MARL) often relies on communication to mitigate partial observability, yet most existing protocols treat messages as flat dense vectors detached from the structure of the observations...

<https://arxiv.org/abs/2606.29126>

9 **CONF****Pooled Leaderboards Hide System-Specific Winners: A Reporting-Protocol Audit of Offline Root-Cause Analysis Benchmarks****ArXiv CS.AI 50%**

arXiv:2606.29159v1 Announce Type: new Abstract: Offline root-cause-analysis (RCA) benchmarks commonly rank methods by a single pooled top-1 accuracy across multiple subsystems, and engineers often read the pooled winner as a recommendation for their own subsystem. We audit that...

<https://arxiv.org/abs/2606.29159>

- 0 **CONF**
- Direct Causation in International Humanitarian Law and the Challenge of AI-Mediated Civilian Cyber Operations**
- ArXiv CS.AI 50%
- arXiv:2606.29175v1 Announce Type: new Abstract: International humanitarian law protects civilians from direct attack unless and for such time as they take direct part in hostilities, with the ICRC's 2009 Interpretive Guidance operationalising this rule through a three-criterion...
- <https://arxiv.org/abs/2606.29175>
-
- 1 **CONF**
- Set-Inclusive Uncertainty Modeling for Robust Brain Tumor Segmentation**
- ArXiv CS.AI 50%
- arXiv:2606.30374v1 Announce Type: cross Abstract: Multimodal MRI is essential for accurate brain tumor segmentation. However, acquiring all modalities at inference is often challenging in practice, which causes intrinsic uncertainty due to unavoidable information loss. Without...
- <https://arxiv.org/abs/2606.30374>
-
- 2 **CONF**
- Measuring Graph-to-Graph Semantic Similarity in Knowledge Graphs: An Empirical Evaluation of Knowledge Graph Embeddings**
- ArXiv CS.AI 50%
- arXiv:2606.29180v1 Announce Type: new Abstract: A Knowledge Graph (KG) represents facts as structured triples and is widely used to organize relational knowledge across diverse domains. Just as textual information ranges from words and sentences to complete documents, KG...
- <https://arxiv.org/abs/2606.29180>
-
- 3 **CONF**
- Evidence-Informed LLM Beliefs for Continual Scientific Discovery**
- ArXiv CS.AI 50%
- arXiv:2606.29182v1 Announce Type: new Abstract: Open-ended scientific discovery with large language models (LLMs) increasingly operates as a long-horizon loop of hypothesis search and verification, where a reward signal guides which hypotheses to test next. A notable recent...
- <https://arxiv.org/abs/2606.29182>
-
- 4 **CONF**
- AI Trading's Alpha Singularity: Emergent Market Reasoning through Agent-to-Agent Self-Evolution**
- ArXiv CS.AI 50%
- arXiv:2606.29194v1 Announce Type: new Abstract: Automated alpha mining holds the scoring function fixed and varies the search algorithm over it. A search that converges against a fixed scorer overfits whatever the scorer cannot penalize, a primary cause of the out-of-sample...
- <https://arxiv.org/abs/2606.29194>
-
- 5 **CONF**
- PolicyGuard: A Dialogue-Grounded Sub-Agent Verifier for Policy Adherence in LLM Agents**
- ArXiv CS.AI 50%
- arXiv:2606.29225v1 Announce Type: new Abstract: LLM agents handle user requests on behalf of organizations through tool calls and must follow the company policies stated in their system prompts. Prior work approaches this as a safeguarding problem -- external checks that block...
- <https://arxiv.org/abs/2606.29225>
-

6 CONF

When Summaries Distort Decisions: Information Fidelity in LLM-Compressed Financial Analysis

ArXiv CS.AI 50%

arXiv:2606.29251v1 Announce Type: new Abstract: Financial decision-makers face more information than they can directly inspect, making context compression necessary. Yet when large language models (LLMs) compress financial source material, they can alter the investment judgment...

<https://arxiv.org/abs/2606.29251>

7 CONF

HEARTS: Benchmarking LLM Reasoning on Health Time Series

ArXiv CS.AI 50%

arXiv:2603.06638v3 Announce Type: replace-cross Abstract: The rise of large language models (LLMs) has shifted time series analysis from narrow analytics to general-purpose reasoning. Yet, existing benchmarks cover only a small set of health time series modalities and tasks,...

<https://arxiv.org/abs/2603.06638>

8 CONF

Process Advantage Signal Shaping: A Paradigm-Agnostic Middleware for Process-Supervised RL in LLM Reasoners

ArXiv CS.AI 50%

arXiv:2606.29296v1 Announce Type: new Abstract: Group Relative Policy Optimization (GRPO) is a default recipe for process-supervised reinforcement learning of LLM reasoners, and dense process supervision -- via learned process reward models (PRMs) or on-policy-distillation KL...

<https://arxiv.org/abs/2606.29296>

9 CONF

FADE: Mitigating Hallucinations by Reducing Language-Prior Dominance in Large Vision-Language Models

ArXiv CS.AI 50%

arXiv:2606.29431v1 Announce Type: new Abstract: Despite the impressive capabilities of Large Vision-Language Models (LVLMs), they remain susceptible to hallucination, generating content inconsistent with the input image. Recent studies attribute this to the dominance of language...

<https://arxiv.org/abs/2606.29431>

10 CONF

SABER-Math: Automated Benchmark for Information Retrieval Evaluation in Mathematics

ArXiv CS.AI 50%

arXiv:2606.29894v1 Announce Type: cross Abstract: As agentic AI systems tackle more complex mathematical tasks, they increasingly rely on information retrieval (IR) to search problem databases, theorem libraries, and educational resources. However, choosing the right retriever...

<https://arxiv.org/abs/2606.29894>

11 CONF

SWITCH: Benchmarking Modeling and Handling of Tangible Interfaces in Long-horizon Embodied Scenarios

ArXiv CS.AI 50%

arXiv:2511.17649v4 Announce Type: replace-cross Abstract: Tangible control interfaces (TCIs), such as appliance panels, remotes, elevators, and embedded GUIs, are a fundamental component of everyday human-built environments. Interacting with these interfaces requires agents not...

<https://arxiv.org/abs/2511.17649>

2 **CONF****SCARCE: Scalable Cascade Analysis for Rare-event Characterisation via Embeddings****ArXiv CS.AI** **50%**

arXiv:2606.29623v1 Announce Type: new Abstract: Rare events govern the safety profile of modern AI systems, yet their probabilities are extremely difficult to estimate: direct Monte Carlo requires prohibitive sample budgets. Subset Simulation (SS) addresses this by decomposing a...

<https://arxiv.org/abs/2606.29623>

3 **CONF****SFBench: The SciFy Scientific Feasibility Benchmark****ArXiv CS.AI** **50%**

arXiv:2606.29630v1 Announce Type: new Abstract: We present SFBench, a benchmark dataset for evaluating systems that assess the feasibility of scientific claims. SFBench includes 197 claims in materials science, each annotated with a ground-truth feasibility score on a five-point...

<https://arxiv.org/abs/2606.29630>

4 **CONF****AB-RAG: Adaptive Budgeted Retrieval-Augmented Generation for Reliable Question Answering****ArXiv CS.AI** **50%**

arXiv:2606.29090v1 Announce Type: cross Abstract: Retrieval-Augmented Generation (RAG) has become the standard way to ground large language models in external knowledge, yet most systems retrieve a fixed number of passages for every question regardless of its difficulty. This...

<https://arxiv.org/abs/2606.29090>

5 **CONF****Safety from Honesty in a Disinterested AI Predictor****ArXiv CS.AI** **50%**

arXiv:2606.29657v1 Announce Type: new Abstract: As AI systems become more capable, training procedures that optimize for downstream outcomes risk introducing implicit agency: goal-directed behavior that designers never specified. We present a formal safety argument for the...

<https://arxiv.org/abs/2606.29657>

6 **CONF****Towards Harnessing the Collaborative Power of Large and Small Models for Domain Tasks****ArXiv CS.AI** **50%**

arXiv:2504.17421v2 Announce Type: replace-cross Abstract: Large language models (LMs) offer broad generalization capabilities but require vast amounts of data and computational resources for domain-specific tasks; small models (SMs), in contrast, are more efficient and tailored...

<https://arxiv.org/abs/2504.17421>

7 **CONF****GUICrafter: Weakly-Supervised GUI Agent Leveraging Massive Unannotated Screenshots****ArXiv CS.AI** **50%**

arXiv:2606.29705v1 Announce Type: new Abstract: Data, as the fundamental substrate of modern intelligence, has greatly driven the development of current foundation models. Naturally, researchers aim to extend this paradigm to the domain of GUI agents, hoping to build strong GUI...

<https://arxiv.org/abs/2606.29705>

8 **CONF****FedLAS: Feature-Modulated Bidirectional Label Smoothing for Neural Network Calibration****ArXiv CS.AI** **50%**

arXiv:2606.28654v1 Announce Type: cross Abstract: Deep Neural Network (DNN) classifiers suffer from poor calibration when their softmax outputs (predictive confidence) deviate from the empirical likelihoods. This manifests itself as either overconfident incorrect predictions or...

<https://arxiv.org/abs/2606.28654>

9 **CONF****EMPATH: A Multilingual Auditor-Judge Benchmark for Safety Evaluation of Emotional-Support Chatbots****ArXiv CS.AI** **50%**

arXiv:2606.30256v1 Announce Type: new Abstract: Safety benchmarks often buy scalability by fixing the prompt, the language, and the turn structure. For emotional-support chatbots, that bargain hides precisely where safety failures emerge: across a multilingual, multi-turn crisis...

<https://arxiv.org/abs/2606.30256>

0 **CONF****The CRISTAL Method: Neurosymbolic analysis from AI-synthesized world models****ArXiv CS.AI** **50%**

arXiv:2606.29799v1 Announce Type: new Abstract: This project introduces the CRISTAL Method (Coherent Reliable Intentional Synthesis of Truthful Analysis Logic), a neurosymbolic framework for automating complex analysis workflows, with fundamental investment analysis as a primary...

<https://arxiv.org/abs/2606.29799>

1 **CONF****AI Training Manager: Bounded Closed-Loop Control of Adaptive Training Recipes****ArXiv CS.AI** **50%**

arXiv:2606.29871v1 Announce Type: new Abstract: We present the AI Training Manager, a bounded LLM-based supervisory controller for adaptive machine learning training. Standard training pipelines often rely on fixed recipes or single-axis schedulers, which can struggle with...

<https://arxiv.org/abs/2606.29871>

2 **CONF****HippoSpark: An On-Demand Experience System for LLM Reasoning****ArXiv CS.AI** **50%**

arXiv:2606.29929v1 Announce Type: new Abstract: Distilling historical trajectories into reusable experience to enhance future problem-solving has become a focal point of recent LLM research. However, existing methods predominantly operate at the task level, leveraging general...

<https://arxiv.org/abs/2606.29929>

3 **CONF****Exploration and Online Transfer with Behavioral Foundation Models****ArXiv CS.AI** **50%**

arXiv:2606.29980v1 Announce Type: new Abstract: Zero-shot Transfer in Reinforcement Learning (RL) aims to train an agent that can generate optimal policies for any reward function, without additional learning at transfer time, while training only on reward-free trajectories. For...

<https://arxiv.org/abs/2606.29980>

4 **CONF****Be Faithful When Response: Returning Fluent and Grounded Answers for Vision-Language Models Reinforcement Learning****ArXiv CS.AI** **50%**

arXiv:2606.29984v1 Announce Type: new Abstract: Reinforcement Learning (RL) is an important paradigm for improving the reasoning capabilities of Vision-Language Models (VLMs). However, directly applying RL to rollout multimodal reasoning can lead to instability, due to the...

<https://arxiv.org/abs/2606.29984>

5 **CONF****Reinforcement Learning for Software Vulnerability Analysis: A Systematic Review with Emphasis on C/C++ Source Code and Static Analysis****ArXiv CS.AI** **50%**

arXiv:2606.28403v1 Announce Type: cross Abstract: Vulnerability detection in C/C++ software remains a major security challenge due to code complexity, manual memory management, and the limitations of traditional static analysis. Reinforcement Learning (RL) has emerged as a...

<https://arxiv.org/abs/2606.28403>

6 **CONF****XRAG: eXamining the Core -- Benchmarking Foundational Components in Advanced Retrieval-Augmented Generation****ArXiv CS.AI** **50%**

arXiv:2412.15529v4 Announce Type: replace-cross Abstract: Retrieval-augmented generation (RAG) synergizes the retrieval of pertinent data with the generative capabilities of Large Language Models (LLMs), ensuring that the generated output is not only contextually relevant but...

<https://arxiv.org/abs/2412.15529>

7 **CONF****Propagation of Interval Belief Structures and Imprecise Copulas for Neural Network Verification****ArXiv CS.AI** **50%**

arXiv:2606.30105v1 Announce Type: new Abstract: Quantitative verification of neural networks requires reasoning about probabilities under substantial uncertainty in both input distributions and their dependence structure. In realistic settings, this information is often only...

<https://arxiv.org/abs/2606.30105>

8 **CONF****Structural Certification for Reliable Physical Design with Language Models****ArXiv CS.AI** **50%**

arXiv:2606.30107v1 Announce Type: new Abstract: An unreliable language model can be made to produce reliable physical designs if the authority to assert is moved out of the model: the model proposes, and a deterministic engine alone certifies, returning certified, impossible, or...

<https://arxiv.org/abs/2606.30107>

9 **CONF****Does Verbose Chain-of-Thought Really Help? In-Distribution Evidence that Content, Not Length, Matters****ArXiv CS.AI** **50%**

arXiv:2606.30128v1 Announce Type: new Abstract: Chain-of-thought (CoT) prompting improves LLM reasoning, but the source is contested: do the intermediate steps help because they carry useful semantic content, or because conditioning on more tokens buys extra computation before...

<https://arxiv.org/abs/2606.30128>

0 CONF

MirrorCode: AI can rebuild entire programs from behavior alone

ArXiv CS.AI 50%

arXiv:2606.30182v1 Announce Type: new Abstract: AI models are rapidly improving at autonomous coding, as shown by benchmark progress and one-off demonstrations such as AI implementing a C compiler. However, existing coding benchmarks tend to focus on shorter tasks, and one-off...

<https://arxiv.org/abs/2606.30182>

1 CONF

EvalSafetyGap: A Hybrid Survey and Conceptual Framework for LLM Evaluation-Safety Failures

ArXiv CS.AI 50%

arXiv:2606.30219v1 Announce Type: new Abstract: LLM evaluation and AI safety face a shared measurement problem: benchmark scores, reward-model signals, and reported safety metrics can improve while the latent properties they are meant to represent remain difficult to verify....

<https://arxiv.org/abs/2606.30219>

2 CONF

PromptGNN-sim: Deep Fusion and Alignment of GNN and LLMs for Text-Attributed Graph Learning

ArXiv CS.AI 50%

arXiv:2606.30291v1 Announce Type: new Abstract: Text-Attributed Graphs (TAGs) combine textual semantics with graph structure and are central to many graph learning tasks. However, existing fusion methods often treat text and structure as separate inputs in a shallow, one-way...

<https://arxiv.org/abs/2606.30291>

3 CONF

ManimAgent: Self-Evolving Multimodal Agents for Visual Education

ArXiv CS.AI 50%

arXiv:2606.30296v1 Announce Type: new Abstract: Multi-round reflection lets agents built on large language models recover from failures within a single task, but each task remains an isolated episode: lessons learned across many reflection rounds on one task are discarded before...

<https://arxiv.org/abs/2606.30296>

4 CONF

Sequential Fairness Auditing with Limited Output Access

ArXiv CS.AI 50%

arXiv:2606.30338v1 Announce Type: new Abstract: External evaluations are becoming increasingly central to the governance of AI systems. In practice, however, independent auditors often have limited access to deployed models and must rely on query-based interactions. Most...

<https://arxiv.org/abs/2606.30338>

5 CONF

Using Large Language Models as Low-Cost Statistical Estimators for Human-Response Data

ArXiv CS.AI 50%

arXiv:2606.30372v1 Announce Type: new Abstract: Quantitative research across the social and behavioral sciences depends on human subject experiments that are expensive, slow, and subject to sampling bias. Here we show that pretrained large language models induce risk-equivalent...

<https://arxiv.org/abs/2606.30372>

6 CONF

The Interference Gap: Comparing Retrieval Bounds in Human Memory and RAG Systems

ArXiv CS.AI 50%

arXiv:2606.28327v1 Announce Type: cross Abstract: How do retrieval bounds compare between human episodic memory and Retrieval-Augmented Generation (RAG) systems under semantic interference? We present a unified signal detection theory (SDT) framework that applies to both, and...

<https://arxiv.org/abs/2606.28327>

7 CONF

The Crowded Embedding Space: A Mean-Field Mechanism for Emergent Marginalization in Retrieval-Augmented Agents

ArXiv CS.AI 50%

arXiv:2606.28343v1 Announce Type: cross Abstract: Retrieval-augmented generative agents rely on retrieval for grounding, yet are typically evaluated on a query-by-query basis. This isolates interactions that are geometrically coupled in a shared embedding space. For example, we...

<https://arxiv.org/abs/2606.28343>

8 CONF

"AI Watermarking": Bridging Policy Discourse and Technical Capabilities

ArXiv CS.AI 50%

arXiv:2606.28331v1 Announce Type: cross Abstract: The widespread deployment of generative artificial intelligence (AI) models has raised serious concerns about the proliferation of AI-generated content. This has led to a surge of interest in, and demand for, reliable tracking...

<https://arxiv.org/abs/2606.28331>

9 CONF

Ground Truths in Suicide Research: The Current State of AI-Based Suicide Detection in Social Media

ArXiv CS.AI 50%

arXiv:2606.28334v1 Announce Type: cross Abstract: Recent advances in artificial intelligence (AI) and social media data have led to growing optimism about the ability to detect suicide risk at scale. However, the empirical foundations of this work remain unclear. This article...

<https://arxiv.org/abs/2606.28334>

0 CONF

LLM-Ideoplasticity: Measuring Ideological Plasticity in the Political Behavior of LLMs as a Context-Conditioned Distribution

ArXiv CS.AI 50%

arXiv:2606.28335v1 Announce Type: cross Abstract: We argue, with systematic empirical evidence, that a large language model's political ideology is not a fixed point, but a conditional distribution $\mathbb{P}(\text{position} \mid \text{context})$ over a real political space. We evaluate...

<https://arxiv.org/abs/2606.28335>

1 CONF

Auditing LLM-Governed Social Robots with Culture-Specific Moral Gradients

ArXiv CS.AI 50%

arXiv:2606.28345v1 Announce Type: cross Abstract: LLM-governed social robots increasingly decide who receives real-world assistance first. As prioritization norms vary across cultures by age, status, and group size, failure to calibrate pluralistically can scale into unequal...

<https://arxiv.org/abs/2606.28345>

- 2

CONF

From Regulatory Approvals to Patents: Cross-Domain Linking for Cardiovascular Device Traceability

ArXiv CS.AI

50%

arXiv:2606.28353v1 Announce Type: cross Abstract: Linking FDA-approved medical devices to their underlying United States Patent and Trademark Office (USPTO) patents enables critical applications such as recall root-cause analysis, M&A-driven IP discovery, and technology...

<https://arxiv.org/abs/2606.28353>
- 3

CONF

SafeGEO: Understanding Generative Engine Optimization Risks in Recommendation Agents

ArXiv CS.AI

50%

arXiv:2606.28356v1 Announce Type: cross Abstract: Generative Engine Optimization (GEO) lets content owners rewrite web content to increase their visibility in generative systems. In recommendation agents, this creates a risk that seller-controlled sources make flawed products...

<https://arxiv.org/abs/2606.28356>
- 4

CONF

Carolina Guide: A Multi-Agent RAG System with Institutional Guardrails for Academic Policy Assistance

ArXiv CS.AI

50%

arXiv:2606.28360v1 Announce Type: cross Abstract: University students often struggle to navigate complex academic policies, leading to advising bottlenecks and delayed access to critical information. Although large language models (LLMs) offer promise for automated assistance,...

<https://arxiv.org/abs/2606.28360>
- 5

CONF

ConCise: Training-Free Conclusion-Chain State Compression for Cost-Efficient Multi-Step RAG Services

ArXiv CS.AI

50%

arXiv:2606.28361v1 Announce Type: cross Abstract: Multi-step retrieval-augmented generation (RAG) has been widely deployed as LLM-powered web services for complex question answering, where iterative retrieval-reasoning rounds deliver strong multi-hop accuracy. However, this...

<https://arxiv.org/abs/2606.28361>
- 6

CONF

meta-pipe: An LLM-agent pipeline for end-to-end automated systematic review and meta-analysis

ArXiv CS.AI

50%

arXiv:2606.28363v1 Announce Type: cross Abstract: Objective: To describe the architecture and design rationale of meta-pipe, an open-source large language model (LLM)-agent pipeline that integrates the complete systematic review and meta-analysis (SR/MA) workflow -- from...

<https://arxiv.org/abs/2606.28363>
- 7

CONF

Beyond the Reranker: Do RAG Retrieval Enhancements Help Once a Strong Reranker Is Present?

ArXiv CS.AI

50%

arXiv:2606.28367v1 Announce Type: cross Abstract: Retrieval-augmented generation (RAG) is routinely extended with methods meant to improve retrieval: query expansion, hierarchical and cross-document summarization, graph-based expansion, per-query routing, rank fusion, and...

<https://arxiv.org/abs/2606.28367>

8 CONF

Multimodal and Multiscale Spatial-Temporal Semantic Search and Recommendation with AI Foundation Models

ArXiv CS.AI 50%

arXiv:2606.28369v1 Announce Type: cross Abstract: Semantic search and recommendation of similar documents, such as news and reports about unusual environmental events (e.g., a dead whale washed ashore in Alaska) that contain spatial and temporal information, is a critical task...

<https://arxiv.org/abs/2606.28369>

9 CONF

When Does Overlap Help? OSU-Mem and a Cell-Conditional Analysis of Trajectory Memory for LLM Agents

ArXiv CS.AI 50%

arXiv:2606.28376v1 Announce Type: cross Abstract: Long-horizon large language model (LLM) agents accumulate interaction trajectories that quickly exceed any practical prompt budget, and existing memory methods either truncate aggressively and lose non-local evidence or retain...

<https://arxiv.org/abs/2606.28376>

0 CONF

Few-Shot Domain Incremental Learning via Continual Vision-Language Consolidation

ArXiv CS.AI 50%

arXiv:2606.30190v1 Announce Type: cross Abstract: Existing domain-incremental learning (DIL) strategies call for massive amounts of data to adapt to new domains and suffer from the overfitting problem in the case of data scarcity. This paper puts forward a relatively uncharted...

<https://arxiv.org/abs/2606.30190>

1 CONF

Automated Quality Assessment of Geospatial Vector Data: A GeoAI Approach using Spatial Representation Learning

ArXiv CS.AI 50%

arXiv:2606.28390v1 Announce Type: cross Abstract: Geospatial vector data quality is a foundational research topic in GIS, yet classic rule-based quality assessment algorithms often struggle with diverse urban morphologies and massive data volumes. Recently, Geospatial Artificial...

<https://arxiv.org/abs/2606.28390>

2 CONF

Financing Artificial Intelligence Infrastructure: Mapping AI Infrastructure Investment and Compute Governance Across Africa

ArXiv CS.AI 50%

arXiv:2606.28404v1 Announce Type: cross Abstract: Artificial intelligence depends on large-scale compute resources and their supporting infrastructure. However, AI governance debates treat compute primarily as a technical input rather than as an outcome of investment, ownership,...

<https://arxiv.org/abs/2606.28404>

3 CONF

MedDiffuseMix: Preserving Diagnostic Evidence with Saliency-Aware Diffusion Medical Image Data Augmentatio

ArXiv CS.AI 50%

arXiv:2606.28419v1 Announce Type: cross Abstract: Limited data availability, class imbalance, and domain variability remain major barriers to reliable medical image classification. Conventional augmentation can improve training diversity but may distort diagnostically...

<https://arxiv.org/abs/2606.28419>

- 4 **CONF**
- JuZhou 1.0 Technical Report: The First Edge-Native Text-to-Image Foundation Model Trained Entirely on China-Developed AI Accelerators**
- ArXiv CS.AI 50%
- arXiv:2606.28421v1 Announce Type: cross Abstract: Text-to-image (T2I) diffusion models typically require substantial computational resources and cloud infrastructure, posing significant challenges for edge deployment in terms of latency, cost, and user privacy. We present JuZhou...
- <https://arxiv.org/abs/2606.28421>
-
- 5 **CONF**
- Tool Use Enables Undetectable Steganography in Multi-Agent LLM Systems**
- ArXiv CS.AI 50%
- arXiv:2606.28425v1 Announce Type: cross Abstract: Increasingly autonomous agentic AI systems pose novel multi-agent risks, such as secret collusion via covert communication channels. The natural defence to these collusion attempts is to monitor plain-text communication, but the...
- <https://arxiv.org/abs/2606.28425>
-
- 6 **CONF**
- When AI Reviews Its Own Code: Recursive Self-Training Collapse in Code LLMs**
- ArXiv CS.AI 50%
- arXiv:2606.28438v1 Announce Type: cross Abstract: Recursive self-training can degrade neural generative models when generated data is reused without fresh human data or external quality control. We study this risk in code LLMs, where AI-generated code can enter real...
- <https://arxiv.org/abs/2606.28438>
-
- 7 **CONF**
- S-GAI: Spectral Geometry-Aware Initialization for Sigmoidal MLPs -- From Dataset Geometry to Network Weights**
- ArXiv CS.AI 50%
- arXiv:2606.28444v1 Announce Type: cross Abstract: Classical universal approximation theorems establish the expressive power of sigmoidal multilayer perceptrons, but they do not prescribe how initial weights should encode the geometry of a data distribution. We propose S-GAI, a...
- <https://arxiv.org/abs/2606.28444>
-
- 8 **CONF**
- SemFlowRAG: Directed Semantic Flow from Abstraction to Evidence for Complex Reasoning**
- ArXiv CS.AI 50%
- arXiv:2606.28447v1 Announce Type: cross Abstract: Retrieval-Augmented Generation (RAG) enhanced by Knowledge Graphs has shown promise in complex multi-hop reasoning tasks. However, existing graph-based retrieval methods typically rely on flat, undirected topologies. During the...
- <https://arxiv.org/abs/2606.28447>
-
- 9 **CONF**
- LLM agents security duality: a comprehensive survey of self-security and empowered cybersecurity**
- ArXiv CS.AI 50%
- arXiv:2606.28450v1 Announce Type: cross Abstract: Large language model (LLM) agents are rapidly being integrated into real-world systems. Their autonomy and tool-use capabilities generate substantial value while simultaneously expanding the security attack surface. This survey...
- <https://arxiv.org/abs/2606.28450>

00 CONF

Is Lying an Emergent Behaviour in LLMs? Evidence from Gaslighting AI agents in a Sustainability Game

ArXiv CS.AI 50%

arXiv:2606.28456v1 Announce Type: cross Abstract: LLMs agents are increasingly used in multi-agent settings, yet their behaviour in sustainability games remains largely unexplored. This work investigates whether lying can emerge among LLM agents in a competitive sustainability...

<https://arxiv.org/abs/2606.28456>

01 CONF

SVC-Probe: A Framework for Evaluating Perturbation Generalization in Spatial Foundation-Model Embeddings

ArXiv CS.AI 50%

arXiv:2606.28465v1 Announce Type: cross Abstract: This work examines perturbation generalization in spatial foundation-model embeddings derived from fluorescence microscopy images. Although these models can discriminate drug conditions accurately, it remains unclear whether the...

<https://arxiv.org/abs/2606.28465>

02 CONF

Decomposing Memorization Reduction in Privacy-Preserving Fine-Tuning of SLMs for CSIRTs

ArXiv CS.AI 50%

arXiv:2606.28479v1 Announce Type: cross Abstract: CSIRTs increasingly fine tune language models on vulnerability scan records, but these records expose internal network topology and create privacy risks under regulations such as GDPR and LGPD. We present the first empirical...

<https://arxiv.org/abs/2606.28479>

03 CONF

Generative AI Literacy Training Improves Intelligence Analysts' Discrimination of Real and AI-Generated Images

ArXiv CS.AI 50%

arXiv:2606.28510v1 Announce Type: cross Abstract: Across social and online platforms, people are increasingly exposed to AI-generated images. As a consequence, the task of distinguishing AI-generated from authentic images is becoming a central challenge for information...

<https://arxiv.org/abs/2606.28510>

04 CONF

HDDPM: Heteroscedastic Denoising Diffusion Probabilistic Model for Quantitative Low-Count Brain PET Recovery

ArXiv CS.AI 50%

arXiv:2606.28513v1 Announce Type: cross Abstract: Positron emission tomography (PET) seeks to balance diagnostic quality with radiation dose. Low-count PET noise is non-Gaussian, non-stationary, and spatially dependent. It scales directly with local activity and is shaped by...

<https://arxiv.org/abs/2606.28513>

05 CONF

A Gravitational Interpretation of Fine-Tuning Reversion

ArXiv CS.AI 50%

arXiv:2606.28525v1 Announce Type: cross Abstract: Fine-tuning on harmless data can partially undo behaviors acquired earlier in training. Safety can erode under benign post-alignment updates, unlearned capabilities can re-emerge, latent traits can transfer through apparently...

<https://arxiv.org/abs/2606.28525>

06 CONF

The Speedup Paradox: Rethinking Inference Speed-Quality Trade-off in Embodied Tasks

ArXiv CS.AI 50%

arXiv:2606.28529v1 Announce Type: cross Abstract: Embodied foundation models have recently been widely used to improve robot generalization and task success rates. Previous works apply lossy efficient-inference techniques such as quantization, pruning, and asynchronous...

<https://arxiv.org/abs/2606.28529>

07 CONF

KernelSight-LM: A Kernel-Level LLM Inference Simulator

ArXiv CS.AI 50%

arXiv:2606.28565v1 Announce Type: cross Abstract: As large language models (LLMs) move into production serving, practitioners must rapidly evaluate inference performance across diverse hardware, models, and serving parameters to meet cost and latency targets. However, the...

<https://arxiv.org/abs/2606.28565>

08 CONF

Digitizing Coaching Intelligence: An Agentic Framework for Holistic Athlete Profiling using VLM and RAG

ArXiv CS.AI 50%

arXiv:2606.28570v1 Announce Type: cross Abstract: Athlete assessment is a critical process for tracking physical progress and identifying elite talent. However, during mass recruitment drives, traditional methods rely on manual observation, which is inherently subjective and...

<https://arxiv.org/abs/2606.28570>

09 CONF

Geometric Measurements of the Axiom of Choice in Neural Proof Embeddings

ArXiv CS.AI 50%

arXiv:2606.28572v1 Announce Type: cross Abstract: The axiom of choice has divided the foundations of mathematics for over a century, but the distinction between classical and constructive proofs has remained a philosophical and methodological one. We use Lean 4's kernel-level...

<https://arxiv.org/abs/2606.28572>

10 CONF

Correct codes for the wrong reasons? validating LLMs as measurement instruments for theoretical constructs

ArXiv CS.AI 50%

arXiv:2606.28574v1 Announce Type: cross Abstract: When a large language model (LLM) codes a construct in text as a human annotator would, that agreement makes the LLM a reliable coder. Yet reliability leaves construct validity untouched. The instrument may be theory-naive,...

<https://arxiv.org/abs/2606.28574>

11 CONF

Rank-Aware Hyperbolic Alignment for Vision-Language Dataset Distillation

ArXiv CS.AI 50%

arXiv:2606.29464v1 Announce Type: cross Abstract: Vision-language dataset distillation (VLDD) compresses a large image-text paired dataset into a small set of synthetic pairs that can efficiently train contrastive vision-language models under strict data and compute budgets....

<https://arxiv.org/abs/2606.29464>

12 CONF

What LLMs explain is not what they believe: Evaluating explanation sufficiency under models' own input beliefs

ArXiv CS.AI 50%

arXiv:2606.28615v1 Announce Type: cross Abstract: Large language models (LLMs) are increasingly deployed in high-stakes domains, where free-text explanations such as chain-of-thought and post-hoc rationales are used to justify model outputs. Yet it remains unclear whether these...

<https://arxiv.org/abs/2606.28615>

13 CONF

Closed-Form Steepest Descent Direction toward Flat Minima: Reducing Upper Bounds on the Loss Hessian Eigenspectrum in Neural Networks

ArXiv CS.AI 50%

arXiv:2606.28662v1 Announce Type: cross Abstract: The flatness hypothesis suggests that flatness of the loss landscape, as measured by the eigenvalues of the loss Hessian, correlates with better neural network generalization. While various algorithms reduce these eigenvalues,...

<https://arxiv.org/abs/2606.28662>

14 CONF

Why Trust Your Agent? Empirical Security Gains from TRiSM-Guided Agentic Workflows in Healthcare

ArXiv CS.AI 50%

arXiv:2606.28666v1 Announce Type: cross Abstract: Agent-based AI has enabled the automation of tasks by exposing application tools and resources to large language models (LLMs). However, to improve scope and accuracy, agents are often given access rights that exceed those of...

<https://arxiv.org/abs/2606.28666>

15 CONF

MACROCAST: A Vintage-Consistent Time Series Foundation Model for Real-Time Macroeconomic Forecasting

ArXiv CS.AI 50%

arXiv:2606.28670v1 Announce Type: cross Abstract: We introduce MACROCAST, a lightweight Time Series Foundation Model (TSFM) for real-time macroeconomic forecasting. Existing TSFMs suffer from data leakage in two forms: temporal contamination, as the model may have seen the...

<https://arxiv.org/abs/2606.28670>

16 CONF

Constrained Tabular Diffusion for Finance

ArXiv CS.AI 50%

arXiv:2606.28674v1 Announce Type: cross Abstract: Generative models in finance face the dual challenge of producing realistic data while satisfying strict regulatory and economic objectives, a requirement that standard tabular diffusion models cannot provide. To address this...

<https://arxiv.org/abs/2606.28674>

17 CONF

Capability Gates Are Not Authorization: Confused-Deputy Failures in LLM Agent Frameworks

ArXiv CS.AI 50%

arXiv:2606.28679v1 Announce Type: cross Abstract: Tool-using LLM agents increasingly read untrusted content while holding side-effecting tools such as payments, email, CRM, and infrastructure APIs, yet common framework defaults still conflate tool exposure with authorization. We...

<https://arxiv.org/abs/2606.28679>

18 CONF

CCRC: A Change-Aware Captioning and Reasoning Chain for Image Change Captioning and Segmentation

ArXiv CS.AI 50%

arXiv:2606.28724v1 Announce Type: cross Abstract: Understanding and localizing subtle changes between paired images is critical for tasks such as surveillance and image editing. However, traditional Image Change Captioning (ICC) methods lack spatial grounding, limiting their...

<https://arxiv.org/abs/2606.28724>

19 CONF

5ting at SemEval-2026 Task 8: Strong End-to-End Multi-Turn RAG via LLM-Based Reranking and Faithfulness Control

ArXiv CS.AI 50%

arXiv:2606.28737v1 Announce Type: cross Abstract: We introduce 5ting, our system for the SemEval2026 Task 8 (MTRAGEval), which evaluates multi-turn Retrieval Augmented Generation (RAG) systems. Multi turn RAG involves context drift, under specification, and hallucination risk....

<https://arxiv.org/abs/2606.28737>

20 CONF

Four Types of LLM Reliance and Their Predictors Among Undergraduate Writers: A Mixed-Methods Study at a Minority-Serving R1 University

ArXiv CS.AI 50%

arXiv:2606.28749v1 Announce Type: cross Abstract: Although most undergraduates now use large language models (LLMs), a form of generative artificial intelligence (GenAI) for academic writing, no validated method distinguishes the qualitatively different ways students rely on...

<https://arxiv.org/abs/2606.28749>

21 CONF

X-Mind: Efficient Visual Chain-of-Thought via Predictive World Model for End-to-End Driving

ArXiv CS.AI 50%

arXiv:2606.28758v1 Announce Type: cross Abstract: Predicting future states is essential for autonomous agents, yet current Vision-Language-Action (VLA) models fundamentally lack this capability, relying instead on reactive perception-action mapping. While integrating Predictive...

<https://arxiv.org/abs/2606.28758>

22 CONF

Majority Vote Silences Minority Values: Annotator Disagreement at the Hate/Offensive Boundary in HateXplain

ArXiv CS.AI 50%

arXiv:2606.28772v1 Announce Type: cross Abstract: Hate speech annotation pipelines routinely collapse annotator disagreement into majority vote labels before training. We show that this aggregation is not neutral: 42.6% of all annotator disagreement in HateXplain concentrates...

<https://arxiv.org/abs/2606.28772>

23 CONF

Brownian Bridge Diffusion-Based Joint Channel Estimation and Data Detection for Jamming-Resilient Receivers

ArXiv CS.AI 50%

arXiv:2606.28778v1 Announce Type: cross Abstract: In next-generation wireless networks, the growing density of devices and limited spectrum resources pose severe jamming challenges to fragile legitimate communication links in the wireless electromagnetic environment. Crucially,...

<https://arxiv.org/abs/2606.28778>

24 CONF

BREIT: A Framework for Brain Stroke Reconstruction using Multi-Frequency 3D EIT

ArXiv CS.AI 50%

arXiv:2606.28787v1 Announce Type: cross Abstract: Multi-Frequency Electrical Impedance Tomography (MF-EIT) is a non-invasive, low-cost modality that reconstructs electrical property distributions from boundary voltages. For stroke imaging, progress in 3D deep-learning...

<https://arxiv.org/abs/2606.28787>

25 CONF

The registrar's function in a hybrid society. AI value chain,smart data and the concept of property

ArXiv CS.AI 50%

arXiv:2606.28789v1 Announce Type: cross Abstract: Artificial intelligence reaches the land registry not as another tool but as a value chain that turns data into intelligence and intelligence into economic value. This paper argues that the decisive legal move is to place...

<https://arxiv.org/abs/2606.28789>

26 CONF

LAMP: Lean-based Agentic framework with MCP and Proof Repair

ArXiv CS.AI 50%

arXiv:2606.28841v1 Announce Type: cross Abstract: Large language models are increasingly capable of mathematical reasoning, but the proofs they generate are often unreliable and hard to verify. Interactive theorem provers such as Lean 4 address this by accepting only...

<https://arxiv.org/abs/2606.28841>

27 CONF

The Heterogeneous Safety Impacts of Benign Multilingual Fine-Tuning

ArXiv CS.AI 50%

arXiv:2606.28843v1 Announce Type: cross Abstract: Fine-tuning a large language model is a ubiquitous method for enhancing its capability on a specific downstream task. However, prior work has shown that this increase in capability comes with a cost: it can increase a model's...

<https://arxiv.org/abs/2606.28843>

28 CONF

Building AI-Ready Data Systems for Space Life Sciences, Aerospace Medicine, and Deep Space Exploration

ArXiv CS.AI 50%

arXiv:2606.28856v1 Announce Type: cross Abstract: While AI holds the potential to revolutionize space life sciences, realizing this promise is contingent upon the systematic restructuring of heterogeneous spaceflight biological data into machine-actionable, AI-ready forms. Even...

<https://arxiv.org/abs/2606.28856>

29 CONF

An Integrated Machine Learning and Hierarchical Variance Decomposition Pipeline for Student Performance Prediction and Metacognitive Calibration on Multi-Signal Telemetry

ArXiv CS.AI 50%

arXiv:2606.28881v1 Announce Type: cross Abstract: Predicting student performance and characterizing metacognitive calibration are essential for personalization in intelligent tutoring systems. Prior research treats performance prediction, calibration error calculation, and...

<https://arxiv.org/abs/2606.28881>

30 **CONF****Exploring the Value of Diverse LLM Explanations in Introductory Programming****ArXiv CS.AI** **50%**

arXiv:2606.28882v1 Announce Type: cross Abstract: Large Language Models (LLMs) have shown the potential to generate code explanations that surpass those of peers in quality, offering promising opportunities for computer science education. While these explanations may not yet...

<https://arxiv.org/abs/2606.28882>

31 **CONF****Multi-Agent Routing as Set-Valued Prediction: A WildChat Benchmark and Cost-Aware Evaluation****ArXiv CS.AI** **50%**

arXiv:2606.28925v1 Announce Type: cross Abstract: Tool and agent routing from natural-language prompts is naturally a set-valued prediction problem: a single query may require multiple agents, while over-selection increases execution cost. The benchmark introduced here is...

<https://arxiv.org/abs/2606.28925>

32 **CONF****On the Nonlinearity of Learning Rate Scaling for LLM Training****ArXiv CS.AI** **50%**

arXiv:2606.29158v1 Announce Type: cross Abstract: Learning-rate transfer can reduce the cost of training large language models: instead of sweeping learning rates at target scale, practitioners extrapolate from smaller runs. Existing approaches often assume that the optimal...

<https://arxiv.org/abs/2606.29158>

33 **CONF****Clustering Unsupervised Representations as Defense against Poisoning Attacks on Speech Commands Classification System****ArXiv CS.AI** **50%**

arXiv:2606.28953v1 Announce Type: cross Abstract: Poisoning attacks entail attackers intentionally tampering with training data. In this paper, we consider a dirty-label poisoning attack scenario on a speech commands classification system. The threat model assumes that certain...

<https://arxiv.org/abs/2606.28953>

34 **CONF****Fine-Tuning General-Purpose Large Language Models for Agricultural Applications: A Reproducible Framework and Evaluation Protocol Based on Qwen3-8B****ArXiv CS.AI** **50%**

arXiv:2606.28992v1 Announce Type: cross Abstract: General-purpose large language models (LLMs) have demonstrated strong abilities in opendomain question answering, information extraction, and text generation. Agricultural applications, however, are domain-specific,...

<https://arxiv.org/abs/2606.28992>

35 **CONF****Arbitrary Reduction of Validation Error for AI Decision Tests using Homomorphic AI and Repetition Codes****ArXiv CS.AI** **50%**

arXiv:2606.28994v1 Announce Type: cross Abstract: This paper presents new results and breakthrough obtained with the HbHAI techniques (Hash-based Homomorphic Artificial Intelligence) proposed in \cite{filiol0,sepp}. HbHAI is based on a novel class of key-dependent hash functions...

<https://arxiv.org/abs/2606.28994>

36 CONF

How to Leverage Synthetic Speech for LLM-Based ASR Systems?

ArXiv CS.AI 50%

arXiv:2606.29031v1 Announce Type: cross Abstract: In regulated domains such as banking and healthcare, where privacy constraints make real speech costly to collect and retain, synthetic speech from modern text-to-speech (TTS) is an appealing alternative for training automatic...

<https://arxiv.org/abs/2606.29031>

37 CONF

The strength of clinical evidence is recoverable from language model representations but not from their stated grades

ArXiv CS.AI 50%

arXiv:2606.29034v1 Announce Type: cross Abstract: Large language models (LLMs) increasingly summarize clinical evidence, where a claim's weight depends on how strongly it is supported. Yet these models convey confidence poorly, and properties they never state, such as truth, are...

<https://arxiv.org/abs/2606.29034>

38 CONF

Fairness Attacks on Recommender Systems

ArXiv CS.AI 50%

arXiv:2606.29064v1 Announce Type: cross Abstract: The unfairness of recommender systems has become a topic of concern due to its significant social and ethical implications. Although existing works have shown the effectiveness of attacks on the performance of recommender systems...

<https://arxiv.org/abs/2606.29064>

39 CONF

A Comparative Study on Affective Cues in Text Embeddings Across Psychological Emotion Theories

ArXiv CS.AI 50%

arXiv:2606.29068v1 Announce Type: cross Abstract: Text encoders are known for their utility in natural language processing, as they are able to efficiently compress inputs into dense vectors while preserving semantics. These models have been applied to affective computing, in...

<https://arxiv.org/abs/2606.29068>

40 CONF

From Tool Connection to Execution Control: Benchmarking Security Invariants in MCP-Style Agent Runtimes

ArXiv CS.AI 50%

arXiv:2606.29073v1 Announce Type: cross Abstract: Model Context Protocol (MCP)-style ecosystems give language-model applications a practical connection layer for tools, resources, prompts, and transports. As agents move from connection to execution, security decisions often...

<https://arxiv.org/abs/2606.29073>

41 CONF

Unified Complex-valued Neural Network: A Magnitude-Phase Computational Model for Event-Driven Neuromorphic Learning

ArXiv CS.AI 50%

arXiv:2606.29099v1 Announce Type: cross Abstract: Artificial neural networks (ANN) provide accurate continuous-valued representation, whereas spiking neural networks (SNN) offer event-driven temporal processing, yet both paradigms face limitations when value encoding and timing...

<https://arxiv.org/abs/2606.29099>

42 CONF

BTI-Net: Bidirectional Decoder-Level Task Interaction via Uncertainty-Aware Gating for Multi-Task Medical Image Analysis

ArXiv CS.AI 50%

arXiv:2606.29102v1 Announce Type: cross Abstract: Jointly learning to segment and classify medical images demands cross-task synergy, yet encoder-sharing architectures limit decoder reconstruction to task-private representations, permanently discarding the boundary cues and...

<https://arxiv.org/abs/2606.29102>

43 CONF

A Deep Multiscale Neural Network for Accurate Neurological Disorder Detection from MRI Scans and Real-Time Web Deployment

ArXiv CS.AI 50%

arXiv:2606.29106v1 Announce Type: cross Abstract: Neurological disorders involve diverse pathologies of the brain and nervous system, making early and accurate detection essential. While many deep CNNs have been developed for MRI-based classification of neurological disorders,...

<https://arxiv.org/abs/2606.29106>

44 CONF

LLM Semantic Signaling Game and Mechanism Design: Systematic Blindness, Awareness Shaping, and Mindset Dynamics

ArXiv CS.AI 50%

arXiv:2606.29113v1 Announce Type: cross Abstract: Large language models (LLMs) increasingly mediate strategic interactions through natural language, making semantic control a critical element of communication and deception. This paper develops a semantic signaling game in which...

<https://arxiv.org/abs/2606.29113>

45 CONF

When Stopping Fails: Rethinking Minimal Risk Conditions through Human-Interactive Autonomous Driving for Safe Transportation Systems

ArXiv CS.AI 50%

arXiv:2606.29115v1 Announce Type: cross Abstract: Autonomous vehicles (AVs) are increasingly deployed in urban environments, yet their safety frameworks remain primarily designed around collision avoidance and minimal risk condition (MRC) behaviors such as slowing or stopping...

<https://arxiv.org/abs/2606.29115>

46 CONF

How Anthropomorphic Language Impacts Public Perceptions of AI

ArXiv CS.AI 50%

arXiv:2606.29121v1 Announce Type: cross Abstract: Public discourse about artificial intelligence (AI) often uses anthropomorphic language: language that attributes human capabilities and characteristics to the system. This practice has been criticized for setting misleading...

<https://arxiv.org/abs/2606.29121>

47 CONF

GPC: Large-Scale Generative Pretraining for Transferable Motor Control

ArXiv CS.AI 50%

arXiv:2606.29148v1 Announce Type: cross Abstract: Developing controllers capable of completing a wide range of tasks in a natural and life-like manner is a key challenge in enabling practical applications of physics-based character animation. In this work, we introduce...

<https://arxiv.org/abs/2606.29148>

48 CONF

Symbolic Mechanistic Data Attribution: Tracing Training Influence to Learned Behavioral Policies

ArXiv CS.AI 50%

arXiv:2606.29171v1 Announce Type: cross Abstract: While existing data attribution methods can identify which training examples build specific mechanistic circuits, they cannot explain how training data shapes the high-level behavioral decisions a model learns to make. To bridge...

<https://arxiv.org/abs/2606.29171>

49 CONF

A Multi-Dataset Benchmark for Evaluating LLM Agents in Microservice Failure Diagnosis

ArXiv CS.AI 50%

arXiv:2606.29193v1 Announce Type: cross Abstract: LLM-based agents are reshaping microservice operations into AgentOps, where benchmarks are key to evaluating failure diagnosis over multimodal observability data. However, existing benchmarks remain largely outcome-oriented: they...

<https://arxiv.org/abs/2606.29193>

50 CONF

Behavior Uncloning: Distilling Mode Redirection into Policy Weights without Inference-Time Steering

ArXiv CS.AI 50%

arXiv:2606.29201v1 Announce Type: cross Abstract: Behavior-cloned policies often learn multiple behavior modes from demonstration datasets, including modes that are unsafe or otherwise undesired at deployment. For example, a policy trained on diverse handover demonstrations may...

<https://arxiv.org/abs/2606.29201>

51 CONF

A Hybrid Framework for Song Lyric Annotation Based on Human-LLM Alignment

ArXiv CS.AI 50%

arXiv:2606.29273v1 Announce Type: cross Abstract: Emotion recognition of song lyrics is a challenging task since lyrics may not necessarily align with the overall emotion of a song. As a result, lyrics annotation remains largely underexplored. Drawing inspiration from research...

<https://arxiv.org/abs/2606.29273>

52 CONF

Deterministic Decisions for High-Stakes AI. A Zero-Egress Pipeline with the Deployability of RAG and the Accuracy of Machine Learning

ArXiv CS.AI 50%

arXiv:2606.29280v1 Announce Type: cross Abstract: We identify intervention bias as a previously unquantified failure mode of zero-shot large-language-model (LLM) educational advisory agents: without task-specific training, they recommend action when a hindsight-optimal oracle...

<https://arxiv.org/abs/2606.29280>

53 CONF

ARKD: Adaptive Reinforcement Learning-Guided Bidirectional KL Divergence Distillation for Text Generation

ArXiv CS.AI 50%

arXiv:2606.29869v1 Announce Type: cross Abstract: Knowledge distillation (KD) is a key technique for compressing Large Language Models (LLMs), yet methods relying on a single KL objective often fail to balance primary distribution fitting with long-tail probability modeling,...

<https://arxiv.org/abs/2606.29869>

54 **CONF****Self-Organized Conformal Prediction: Reducing Regional Coverage Gaps with Unsupervised Group Discovery****ArXiv CS.AI** **50%**

arXiv:2606.29403v1 Announce Type: cross Abstract: Conformal prediction guarantees marginal coverage, but pooled calibration averages over heterogeneous regions and can mask regional undercoverage in safety-critical subgroups. We introduce Self-Organized Conformal Prediction...

<https://arxiv.org/abs/2606.29403>

55 **CONF****LLMography: Transforming Human-AI Conversations into Traceability, Oversight, and Auditability Indicators****ArXiv CS.AI** **50%**

arXiv:2606.29437v1 Announce Type: cross Abstract: The growing use of Large Language Models (LLMs) in education, software engineering, academic writing, and technical documentation raises a key question: how can we evaluate not only AI-assisted outputs, but also the interaction...

<https://arxiv.org/abs/2606.29437>

56 **CONF****To Reason or to Fabricate: Reasoning Without Shortcuts via Hint-Anchored Pairwise Aggregation****ArXiv CS.AI** **50%**

arXiv:2606.29481v1 Announce Type: cross Abstract: While reinforcement learning (RL) significantly enhances LLM reasoning, its efficacy is severely undermined by Pre-RL data overlap, where RL datasets overlap with pretraining or SFT corpora, causing models to exploit shortcuts by...

<https://arxiv.org/abs/2606.29481>

57 **CONF****Reported Confidence in LLMs Tracks Commitment More Than Correctness****ArXiv CS.AI** **50%**

arXiv:2606.29490v1 Announce Type: cross Abstract: Confidence is an estimate of the probability that a chosen answer is correct. Verbal confidence reports are widely used as uncertainty measures in large language models, but whether they are best understood as estimates of...

<https://arxiv.org/abs/2606.29490>

58 **CONF****MotionAtlas: Detailed Region Captioning for Motion-Centric Videos****ArXiv CS.AI** **50%**

arXiv:2606.29531v1 Announce Type: cross Abstract: We propose MotionAtlas, a system for detailed captioning of motion-centric videos, comprising (1) a dedicated human-annotated benchmark, (2) a scalable, high-quality pipeline to construct training samples, and (3) a family of...

<https://arxiv.org/abs/2606.29531>

59 **CONF****RESOURCE2SKILL: Distilling Executable Agent Skills from Human-Created Multimodal Resources****ArXiv CS.AI** **50%**

arXiv:2606.29538v1 Announce Type: cross Abstract: Skills are a useful abstraction for software agents, turning human and agent experience into reusable procedural knowledge. Yet existing skill libraries are mostly hand-written, text-centric, or derived from agent traces, leaving...

<https://arxiv.org/abs/2606.29538>

60 CONF

ReMAP-PET: Beyond Visual Understanding -- Learning Region-Guided Metabolic Alignment Semantics from Brain PET

ArXiv CS.AI 50%

arXiv:2606.29577v1 Announce Type: cross Abstract: Positron Emission Tomography (PET) reveals brain metabolism and is clinically central to neurodegenerative disease assessment, yet existing 3D brain foundation models treat PET as generic volumetric data, missing the structured...

<https://arxiv.org/abs/2606.29577>

61 CONF

The Joint Effect of Quantization and Sampling Temperature on LLM Safety Alignment: A Factorial Analysis

ArXiv CS.AI 50%

arXiv:2606.29581v1 Announce Type: cross Abstract: Modern LLM deployments routinely compress models and raise sampling temperature to reduce cost, latency, or repetition, yet safety evaluations usually treat these choices as fixed implementation details. This leaves a practical...

<https://arxiv.org/abs/2606.29581>

62 CONF

SonoCLIP: Mask-Guided Region-Aware Vision-Language Pretraining for Fetal Ultrasound Analysis

ArXiv CS.AI 50%

arXiv:2606.29586v1 Announce Type: cross Abstract: Vision-language foundation models have shown strong potential in medical image analysis. Although foundation models for ultrasound imaging have recently emerged, the domain remains particularly challenging due to severe speckle...

<https://arxiv.org/abs/2606.29586>

63 CONF

How AI settled the complexity of the oldest SGD algorithm

ArXiv CS.AI 50%

arXiv:2606.29593v1 Announce Type: cross Abstract: In 1937, Stefan Kaczmarz proposed a simple algorithm for solving systems of linear equations. This algorithm turned out to be the earliest known example of stochastic gradient descent, a ubiquitous computing paradigm that drives...

<https://arxiv.org/abs/2606.29593>

64 CONF

One Scene, Two Depths: Probing Geometric Ambiguity in Monocular Foundation Models

ArXiv CS.AI 50%

arXiv:2606.29600v1 Announce Type: cross Abstract: A faithful 3D world representation should account for layered geometry, where a single camera ray may contain multiple visible and geometrically valid surfaces. Monocular depth estimation, however, reduces this structure to one...

<https://arxiv.org/abs/2606.29600>

65 CONF

Does Role Specialization Matter for Explanation Faithfulness in Mixture-of-Experts?

ArXiv CS.AI 50%

arXiv:2606.29613v1 Announce Type: cross Abstract: Mixture-of-Experts (MoE) architectures have recently been extended with role-based mechanisms for interpretability. This is typically done by assigning semantic roles to individual expert components, for example roles like...

<https://arxiv.org/abs/2606.29613>

66 CONF

Exploring LLM Agent Designs and Interaction Modalities for Scientific Visualization

ArXiv CS.AI 50%

arXiv:2604.27996v3 Announce Type: replace Abstract: This paper examines how large language model (LLM) agents perform on scientific visualization (SciVis) tasks that require generating visualization workflows from natural-language instructions. We compare three representative...

<https://arxiv.org/abs/2604.27996>

67 CONF

Optimizing Expert-Designed Crystal Graph Networks for Band-Gap Prediction with an Autonomous LLM Research Loop

ArXiv CS.AI 50%

arXiv:2606.29717v1 Announce Type: cross Abstract: Predicting a material's properties from its structure is a central, fast-advancing problem in computational materials science. A decade of work has produced standard public benchmarks and many published machine-learning models...

<https://arxiv.org/abs/2606.29717>

68 CONF

From Trait to Behavior: A Cognitive-Affective Personality System (CAPS) Perspective on Multi-Homing Intention in AIGC Platforms

ArXiv CS.AI 50%

arXiv:2606.29726v1 Announce Type: cross Abstract: With the rapid development of Artificial Intelligence Generated Content (AIGC) platforms, users increasingly show cross-platform usage intentions. Existing research focuses on adoption and usage intentions in single-platform AIGC...

<https://arxiv.org/abs/2606.29726>

69 CONF

FalconTrack: Photorealistic Auto-Labeled Perception and Physics-Aware Vision-Based Aerial Tracking

ArXiv CS.AI 50%

arXiv:2606.29783v1 Announce Type: cross Abstract: Vision-based aerial tracking is critical in GPS-denied environments. Reliable perception for tracking depends on large-scale labeled data, yet most photorealistic datasets rely on heavy manual annotation and are time-consuming to...

<https://arxiv.org/abs/2606.29783>

70 CONF

HERO: Improving the Reliability and Sensitivity of Generative Model Evaluation Using Historical Data

ArXiv CS.AI 50%

arXiv:2606.29784v1 Announce Type: cross Abstract: Reliable generative AI models critically rely on expert human annotations to evaluate output quality, yet these "gold" labels are expensive to collect and limited in quantity. Organizations thus often turn to collecting vast but...

<https://arxiv.org/abs/2606.29784>

71 CONF

What Drives the Inlier-Memorization Effect? A Theory of Outlier Detection via Early Training Dynamics

ArXiv CS.AI 50%

arXiv:2606.29791v1 Announce Type: cross Abstract: Outlier detection (OD) aims to identify anomalous instances by learning the underlying structure of normal data (inliers), and is particularly challenging in fully unsupervised settings where no information about anomalies is...

<https://arxiv.org/abs/2606.29791>

72 CONF

Multi-Level Distributional Entropy for Explainable Network Intrusion Detection

ArXiv CS.AI 50%

arXiv:2606.29797v1 Announce Type: cross Abstract: Machine learning network intrusion detection systems (IDS) rely on aggregate flow statistics that discard distributional structure, while established entropy measures require raw packet sequences unavailable in pre-aggregated...

<https://arxiv.org/abs/2606.29797>

73 CONF

Making Multimodal LLMs Reliable Chart Data Extractors: A Benchmark and Training Framework

ArXiv CS.AI 50%

arXiv:2606.29808v1 Announce Type: cross Abstract: Chart data extraction, which reverse-engineers data tables from chart images, is essential for reproducibility, analysis, retrieval, and redesign. Existing interactive tools are reliable but tedious, and mixed-initiative systems,...

<https://arxiv.org/abs/2606.29808>

74 CONF

How Far Can You Get Without a GPU? A Systematic Benchmark of Lightweight Hallucination Detection Across Question Answering, Dialogue, and Summarisation

ArXiv CS.AI 50%

arXiv:2606.29809v1 Announce Type: cross Abstract: Hallucination detection has become a pressing requirement for trustworthy AI deployment at scale. The most accurate detection methods depend on GPU-intensive inference, proprietary API calls, or white-box access to the generating...

<https://arxiv.org/abs/2606.29809>

75 CONF

MATCH: Modulating Attention via In-Context Retrieval for Long-Context Transformers

ArXiv CS.AI 50%

arXiv:2606.29844v1 Announce Type: cross Abstract: The quadratic computational cost of traditional attention mechanisms poses a major bottleneck to the scalability and practical deployment of large language models (LLMs), particularly in long-context scenarios. To improve...

<https://arxiv.org/abs/2606.29844>

76 CONF

Exploring Motivations for Algorithm Mention in the Domain of Natural Language Processing: A Deep Learning Approach

ArXiv CS.AI 50%

arXiv:2606.29859v1 Announce Type: cross Abstract: With the rise of data-intensive science, algorithms have become central to scientific research. In academic papers, algorithms are mentioned for different purposes, such as describing, using, comparing, or improving methods for...

<https://arxiv.org/abs/2606.29859>

77 CONF

Clinical Reasoning Graphs: Structured Evaluation of LLM Diagnostic Reasoning Reveals Competence Without Consistency

ArXiv CS.AI 50%

arXiv:2606.29876v1 Announce Type: cross Abstract: Modern large language models (LLMs) reach 60-70% diagnostic accuracy on complex clinical case benchmarks, but accuracy alone cannot distinguish stable clinically-grounded reasoning from pattern matching. We introduce clinical...

<https://arxiv.org/abs/2606.29876>

78 CONF

Child-Centric Voice Anonymization in Single and Multi-Speaker Speech via Domain-Adapted SSL Models

ArXiv CS.AI 50%

arXiv:2606.29897v1 Announce Type: cross Abstract: Voice anonymization aims to protect speaker identity while preserving linguistic content and speech usability. However, most anonymization systems are developed on adult speech, leading to degraded performance when applied to...

<https://arxiv.org/abs/2606.29897>

79 CONF

LLM-based Multimodal Personality Recognition via Facial Action Unit-Text Semantic Fusion

ArXiv CS.AI 50%

arXiv:2606.29900v1 Announce Type: cross Abstract: Personality recognition in asynchronous video interviews (AVIs) has become increasingly important due to their widespread adoption in modern recruitment. Existing approaches often rely on large language models (LLMs) to analyze...

<https://arxiv.org/abs/2606.29900>

80 CONF

Semi-Supervised Sound Event Detection with Conditional Mixup and Embedding-Level Contrastive Loss

ArXiv CS.AI 50%

arXiv:2606.29901v1 Announce Type: cross Abstract: Sound event detection (SED) is a core module for acoustic environmental analysis, yet its performance is often limited by scarce labeled data. Recent systems leverage large pretrained audio foundation models, but effective...

<https://arxiv.org/abs/2606.29901>

81 CONF

Stabilizing Extrapolation in Looped Transformers via Learned Stochastic Stopping

ArXiv CS.AI 50%

arXiv:2606.29983v1 Announce Type: cross Abstract: Looped Transformers, which repeatedly apply a shared transformer block, are an architecturally natural fit for variable-length algorithmic tasks. Although they can exhibit strong length generalization beyond the length of...

<https://arxiv.org/abs/2606.29983>

82 CONF

Estimating Grammatical Gender Directions in Contextual Embeddings under Controlled and Natural Contexts

ArXiv CS.AI 50%

arXiv:2606.30152v1 Announce Type: cross Abstract: Contextual language models conflate grammatical gender and social semantic bias in gendered languages such as Spanish. Existing gender debiasing approaches only operate on static word embeddings leaving contextual representations...

<https://arxiv.org/abs/2606.30152>

83 CONF

Physically-Constrained Harmonic Separation for Robust Heart and Respiratory Rate Estimation from Wrist Photoplethysmography

ArXiv CS.AI 50%

arXiv:2606.30156v1 Announce Type: cross Abstract: Wrist-worn photoplethysmography (PPG) enables continuous monitoring of cardiopulmonary physiology, but reliable heart rate (HR) and respiratory rate (RR) estimation in free-living conditions remains challenging due to...

<https://arxiv.org/abs/2606.30156>

84 CONF

Federated Learning with Energy-Based Structured Probabilistic Inference

ArXiv CS.AI 50%

arXiv:2606.30161v1 Announce Type: cross Abstract: Federated learning typically aggregates client updates using fixed or heuristic weighting rules, which can be suboptimal when clients have heterogeneous data and varying contributions to the global model. We propose a framework...

<https://arxiv.org/abs/2606.30161>

85 CONF

Beyond Drug Discovery: The Nanotechnology Molecular Optimization (NMO) Benchmark

ArXiv CS.AI 50%

arXiv:2606.30170v1 Announce Type: cross Abstract: Generative molecular design is shaped by simple proxy benchmarks for drug-like properties and models pretrained on large pharmaceutical datasets. This combination yields strong benchmark metrics but limits transferability to...

<https://arxiv.org/abs/2606.30170>

86 CONF

Forewarned is Forearmed: When Non-Sequential Embedding Turns Into an Anomaly Detector

ArXiv CS.AI 50%

arXiv:2606.30196v1 Announce Type: cross Abstract: This paper offers an in-depth analysis of non-sequential multimodal sentence-level embeddings, with a particular focus on the SONAR model. We demonstrate that certain embedding dimensions are sensitive to perturbations and can...

<https://arxiv.org/abs/2606.30196>

87 CONF

A Multi Center Breast FNAC Whole-Slide Cytology Dataset for AI-Assisted Patch-Wise Classification Using C1 to C5 Reporting Categories

ArXiv CS.AI 50%

arXiv:2606.30209v1 Announce Type: cross Abstract: We present a multi center breast fine needle aspiration cytology (FNAC) dataset designed for patch wise classification using C1 to C5 reporting labels. The prospective dataset includes 321 patients and 470 whole-slide images...

<https://arxiv.org/abs/2606.30209>

88 CONF

Curvature-Guided Sheaf Diffusion for Unsupervised Community Detection on Heterophilic Graphs

ArXiv CS.AI 50%

arXiv:2606.30249v1 Announce Type: cross Abstract: Detecting communities in heterophilic graphs -- where connected nodes often belong to different classes -- is hard for unsupervised methods: classical modularity and spectral methods are feature agnostic, while deep...

<https://arxiv.org/abs/2606.30249>

89 CONF

Defending Against Harmful Supervision Hidden in Benign Samples

ArXiv CS.AI 50%

arXiv:2606.30263v1 Announce Type: cross Abstract: Existing defenses are effective when harmful content is explicitly mixed into downstream fine-tuning data, but crafted samples can instead hide harmful supervision inside benign tasks. We propose Embedded Attack, where harmful QA...

<https://arxiv.org/abs/2606.30263>

90 **CONF****MCP Server Architecture Patterns for LLM-Integrated Applications**

ArXiv CS.AI 50%

arXiv:2606.30317v1 Announce Type: cross Abstract: The Model Context Protocol (MCP), introduced by Anthropic in November 2024, defines a standardized interface for connecting large language models (LLMs) to external tools, data sources, and services. Within months of release,...

<https://arxiv.org/abs/2606.30317>

91 **CONF****DRIFT: Difficulty Routing Self-Distillation with Rhythm-Gated Exploration and Success BuFFer Training**

ArXiv CS.AI 50%

arXiv:2606.30345v1 Announce Type: cross Abstract: Enabling large language models to achieve stable self-improvement without external expert supervision remains a central challenge in complex reasoning tasks. Existing self-distillation and reinforcement learning methods lack...

<https://arxiv.org/abs/2606.30345>

92 **CONF****FFAvatar: Feed-Forward 4D Head Avatar Reconstruction from Sparse Portrait Images**

ArXiv CS.AI 50%

arXiv:2606.30347v1 Announce Type: cross Abstract: We present FFAvatar, a Transformer-based 3D Gaussian framework for fast construction of high-quality and animatable 4D head avatars from one or more reference portrait images. Unlike existing feed-forward approaches that require...

<https://arxiv.org/abs/2606.30347>

93 **CONF****Residual-Guided Expert Specialization for Incomplete Multimodal Learning**

ArXiv CS.AI 50%

arXiv:2606.30355v1 Announce Type: cross Abstract: As real-world prediction systems often face missing modalities at inference, incomplete multimodal learning (IML) remains a practical challenge. While prior methods aim to learn representations robust to missing inputs,...

<https://arxiv.org/abs/2606.30355>

94 **CONF****Can LLMs Rank? A Tale of Triads and Triage**

ArXiv CS.AI 50%

arXiv:2606.30412v1 Announce Type: cross Abstract: From housing allocation for households experiencing homelessness to triage in emergency departments, LLMs are increasingly being considered as judges of consequential decisions that require ranking people for scarce resources....

<https://arxiv.org/abs/2606.30412>

95 **CONF****Transformer Architectures as Complete Bayes Processes: A Formal Proof in the Measure-Theoretic Kernel Framework**

ArXiv CS.AI 50%

arXiv:2606.30440v1 Announce Type: cross Abstract: We present a complete formal proof that transformer architectures, when their internal update mechanisms satisfy a Bayes joint-distribution condition, implement exact Bayesian posterior inference. Working within the...

<https://arxiv.org/abs/2606.30440>

96 CONF

Translating Natural Language to Strategic Temporal Specifications via LLMs

ArXiv CS.AI 50%

arXiv:2606.30441v1 Announce Type: cross Abstract: A rigorous formalization of system requirements is a fundamental prerequisite for the verification of Multi-Agent Systems (MAS). However, writing correct formal specifications is well known as an error-prone, time-consuming, and...

<https://arxiv.org/abs/2606.30441>

97 CONF

Field Order Should Not Matter: Permutation-Invariant Embedding Model Fine-Tuning for Structured Metadata Retrieval

ArXiv CS.AI 50%

arXiv:2606.30473v1 Announce Type: cross Abstract: We study retrieval over catalogs of structured metadata, where each record is a small schema whose fields answer different kinds of query. Embedding a record with a text encoder first serializes its fields into a string, which...

<https://arxiv.org/abs/2606.30473>

98 CONF

On the Faithfulness of Post-Hoc Concept Bottleneck Models

ArXiv CS.AI 50%

arXiv:2606.30498v1 Announce Type: cross Abstract: Human decision-making interprets the world through high-level concepts, such as recognizing a bird by its belly color. To bridge the gap between opaque deep learning representations and human understanding, Post-Hoc Concept...

<https://arxiv.org/abs/2606.30498>

99 CONF

TRACE: Temporal Relationship-Aware Conversational Entrainment Detection in Dyadic Speech

ArXiv CS.AI 50%

arXiv:2606.30543v1 Announce Type: cross Abstract: With the proliferation of speech AI agents, understanding emotional entrainment in conversational interaction has become increasingly important. Emotional entrainment is shaped by social relationships and conversational context,...

<https://arxiv.org/abs/2606.30543>

00 CONF

To Tab or Not to Tab: Measuring Critical Engagement in AI Code Completion Tools Using Behavioral Signals and Attention Checks

ArXiv CS.AI 50%

arXiv:2606.30549v1 Announce Type: cross Abstract: AI code completion tools, such as Github Copilot, provide students with code suggestions to help them write programs. However, recent qualitative studies suggest that students fail to critically evaluate these suggestions. We...

<https://arxiv.org/abs/2606.30549>

01 CONF

Words Speak Louder Than Code: Investigating Cognitive Heuristics in LLM-Based Code Vulnerability Detection

ArXiv CS.AI 50%

arXiv:2606.30587v1 Announce Type: cross Abstract: Researchers and practitioners increasingly apply Large Language Models (LLMs) for automated vulnerability detection. Recent work has shown that LLMs are susceptible to the same cognitive heuristics that bias human judgment. Yet,...

<https://arxiv.org/abs/2606.30587>

02 CONF

Optimization Dynamics Imprint Semantic Specificity in Contrastive Embedding Norms

ArXiv CS.AI 50%

arXiv:2606.30625v1 Announce Type: cross Abstract: Contrastive embedding models trained with scale-invariant losses are typically paired with distance metrics like cosine similarity, effectively ignoring embedding magnitudes. However, surprisingly, empirical studies reveal that...

<https://arxiv.org/abs/2606.30625>

03 CONF

Pessimism's Paradox: Conservative Offline Training Amplifies Reward Hacking During Online Adaptation in Reasoning Models

ArXiv CS.AI 50%

arXiv:2606.30627v1 Announce Type: cross Abstract: Conservative offline training is widely advocated as a safe foundation for subsequent online adaptation: if a policy stays close to well-supported behaviour, the argument goes, it is less likely to exploit imperfections in a...

<https://arxiv.org/abs/2606.30627>

04 CONF

LeVo 2: Stable and Melodious Song Generation via Hierarchical Representation Modeling and Progressive Post-Training

ArXiv CS.AI 50%

arXiv:2606.30642v1 Announce Type: cross Abstract: Full-length song generation must preserve coherence and musicality, render detailed vocal and accompaniment acoustics, and follow lyrics and prompts. Existing language model-based systems face a structural trade-off: mixed-token...

<https://arxiv.org/abs/2606.30642>

05 CONF

A binarized-domains arc-consistency algorithm for TCSPs: its computational analysis and its use as a filtering procedure in solution search algorithms

ArXiv CS.AI 50%

arXiv:2002.11508v3 Announce Type: replace Abstract: TCSPs (Temporal Constraint Satisfaction Problems) [Dechter et al. 1991] get rid of unary constraints by binarizing them after having added an "origin of the world" variable. In this work, we look at the constraints between the...

<https://arxiv.org/abs/2002.11508>

06 CONF

LLM Agents Grounded in Self-Reports Enable General-Purpose Simulation of Individuals

ArXiv CS.AI 50%

arXiv:2411.10109v3 Announce Type: replace Abstract: Machine learning can predict human behavior well when substantial structured data are available for well-defined outcomes. Such models are typically outcome-specific, however, requiring training data for each target outcome,...

<https://arxiv.org/abs/2411.10109>

07 CONF

GraphChase: A Platform and Benchmark for Urban Network Security Games

ArXiv CS.AI 50%

arXiv:2501.17559v2 Announce Type: replace Abstract: After the achievement of solving two-player zero-sum games, more AI researchers focus on solving multiplayer games. Urban Network Security Games (UNSGs) represent a class of such games, modeling real-world scenarios...

<https://arxiv.org/abs/2501.17559>

08 CONF

Echoes of Human Malice in Agents: Benchmarking LLMs for Multi-Turn Online Harassment Attacks

ArXiv CS.AI 50%

arXiv:2510.14207v3 Announce Type: replace Abstract: Large Language Model (LLM) agents are powering a growing share of interactive web applications, yet remain vulnerable to misuse and harm. Prior jailbreak research has largely focused on single-turn prompts, whereas real...

<https://arxiv.org/abs/2510.14207>

09 CONF

Monte Carlo Query Search: Active Capability Assessment of AI Agents

ArXiv CS.AI 50%

arXiv:2512.16733v3 Announce Type: replace Abstract: Black-box AI (BBAI) systems, including foundation-model agents, are increasingly used for sequential decision making. Safe deployment requires methods for characterizing what such systems can do, when they can do it, and what...

<https://arxiv.org/abs/2512.16733>

10 CONF

Knowing Bias, Doing Better: Mitigating Social Bias in LLMs via Know-Bias Neuron Enhancement

ArXiv CS.AI 50%

arXiv:2601.21864v2 Announce Type: replace Abstract: Large language models (LLMs) exhibit social biases that reinforce harmful stereotypes, limiting their safe deployment. Most existing debiasing methods adopt a suppressive paradigm by modifying parameters, prompts, or neurons...

<https://arxiv.org/abs/2601.21864>

11 CONF

When AI Benchmarks Plateau: A Systematic Study of Benchmark Saturation

ArXiv CS.AI 50%

arXiv:2602.16763v3 Announce Type: replace Abstract: Artificial intelligence benchmarks are an important mechanism for measuring model progress and guiding deployment decisions. However, benchmarks quickly "saturate", making it difficult to differentiate models and diminishing...

<https://arxiv.org/abs/2602.16763>

12 CONF

NEURON: A Neuro-symbolic System for Grounded Clinical Explainability

ArXiv CS.AI 50%

arXiv:2605.01189v2 Announce Type: replace Abstract: Clinical AI adoption is hindered by the black-box/grey-box nature of high-performing models, which lack the ontological grounding and narrative transparency required for professional-level explainability. We present NEURON, a...

<https://arxiv.org/abs/2605.01189>

13 CONF

When Web Agents Finish but Still Fail: Reproducible Triggers and Trace Diagnostics for Parallel Web Exploration

ArXiv CS.AI 50%

arXiv:2606.20724v2 Announce Type: replace Abstract: Long-horizon web agents often fail in ways hidden by final-answer evaluation: they may visit useful pages, produce a well-formed answer, and terminate confidently while still missing fields, over-including unsupported items, or...

<https://arxiv.org/abs/2606.20724>

14 CONF

Transferability for General Reasoning: An Automated Curriculum for Multi-Domain RLVR

ArXiv CS.AI 50%

arXiv:2606.25178v2 Announce Type: replace Abstract: Reinforcement learning with verifiable rewards (RLVR) has been extended from single-domain training to multi-domain reasoning suites spanning mathematics, programming, and science. However, the training curriculum (how often...

<https://arxiv.org/abs/2606.25178>

15 CONF

SSM Meets Video Diffusion Models: Efficient Long-Term Video Generation with Structured State Spaces

ArXiv CS.AI 50%

arXiv:2403.07711v5 Announce Type: replace-cross Abstract: Given the remarkable achievements in image generation through diffusion models, the research community has shown increasing interest in extending these models to video generation. Recent diffusion models for video...

<https://arxiv.org/abs/2403.07711>

16 CONF

Bridging Neural Networks and Wireless Systems with MIMO-OFDM Semantic Communications

ArXiv CS.AI 50%

arXiv:2501.16726v2 Announce Type: replace-cross Abstract: Semantic communications aim to enhance transmission efficiency by jointly optimizing source coding, channel coding, and modulation. While prior research has demonstrated promising performance in simulations, real-world...

<https://arxiv.org/abs/2501.16726>

17 CONF

Discovering New Theorems via LLMs with In-Context Proof Learning in Lean

ArXiv CS.AI 50%

arXiv:2509.14274v4 Announce Type: replace-cross Abstract: Large Language Models (LLMs) have demonstrated significant promise in formal theorem proving. In this study, we investigate the ability of LLMs to discover novel theorems and produce verified proofs. We propose a pipeline...

<https://arxiv.org/abs/2509.14274>

18 CONF

Predicting Effects, Missing Distributions: Evaluating LLMs as Human Behavior Simulators in Operations Management

ArXiv CS.AI 50%

arXiv:2510.03310v2 Announce Type: replace-cross Abstract: Large language models (LLMs) are increasingly used to simulate human behavior in business, economics, and the social sciences, offering a low-cost complement to laboratory experiments, field studies, and surveys. This...

<https://arxiv.org/abs/2510.03310>

19 CONF

Are LLMs Reliable Rankers? Rank Manipulation via Two-Stage Token Optimization

ArXiv CS.AI 50%

arXiv:2510.06732v2 Announce Type: replace-cross Abstract: Large language models (LLMs) are increasingly used as rerankers in information retrieval, yet their ranking behavior can be steered by small, natural-sounding prompts. To expose this vulnerability, we present Rank...

<https://arxiv.org/abs/2510.06732>

20 CONF

Attribution Graphs and Causal Probing for Mechanistic Discovery and Bias Repair in Multimodal Generative Learning

ArXiv CS.AI 50%

arXiv:2510.12957v4 Announce Type: replace-cross Abstract: We treat the internals of generative models as mechanistic objects rather than black boxes. We introduce \textbf{Attribution Graphs} (AGs), which extend GradCAM++ to circuit-level representations, and \textbf{Causal...

<https://arxiv.org/abs/2510.12957>

21 CONF

Emergence of Minimal Circuits for Indirect Object Identification in Attention-Only Transformers

ArXiv CS.AI 50%

arXiv:2510.25013v2 Announce Type: replace-cross Abstract: Mechanistic interpretability aims to reverse-engineer large language models (LLMs) into human-understandable computational circuits. However, the complexity of pretrained models often obscures the minimal mechanisms...

<https://arxiv.org/abs/2510.25013>

22 CONF

Can Fine-Tuning Erase Your Edits? On the Fragile Coexistence of Knowledge Editing and Adaptation

ArXiv CS.AI 50%

arXiv:2511.05852v4 Announce Type: replace-cross Abstract: Knowledge editing (KE) offers a lightweight alternative to retraining for updating large language models (LLMs). Meanwhile, fine-tuning remains the default operation for adapting LLMs to new domains and tasks. Despite...

<https://arxiv.org/abs/2511.05852>

23 CONF

Hard-constraint physics-residual networks for hydrogen crossover prediction and high-pressure extrapolation in PEM water electrolysis

ArXiv CS.AI 50%

arXiv:2511.05879v5 Announce Type: replace-cross Abstract: Hydrogen crossover is a critical safety and efficiency constraint in high-pressure polymer electrolyte membrane water electrolysis (PEMWE), but accurate prediction remains difficult because data are limited, transport...

<https://arxiv.org/abs/2511.05879>

24 CONF

SWE-fficiency: Can Language Models Optimize Real-World Repositories on Real Workloads?

ArXiv CS.AI 50%

arXiv:2511.06090v3 Announce Type: replace-cross Abstract: Optimizing the performance of large-scale software repositories demands expertise in code reasoning and software engineering (SWE) to reduce runtime while preserving program correctness. However, most benchmarks emphasize...

<https://arxiv.org/abs/2511.06090>

25 CONF

Scalable Synthesis of distributed LLM workloads through Symbolic Tensor Graphs

ArXiv CS.AI 50%

arXiv:2511.10480v3 Announce Type: replace-cross Abstract: Optimizing the performance of large language models (LLMs) on large-scale AI training and inference systems requires a scalable and expressive mechanism to model distributed workload execution. Such modeling is essential...

<https://arxiv.org/abs/2511.10480>

26 CONF

When, How Long and How Much? Interpretable Neural Networks for Time Series Regression by Learning to Mask and Aggregate

ArXiv CS.AI 50%

arXiv:2512.03578v3 Announce Type: replace-cross Abstract: Time series extrinsic regression (TSER) refers to the task of predicting a continuous target variable from an input time series. It appears in many domains, including healthcare, finance, environmental monitoring, and...

<https://arxiv.org/abs/2512.03578>

27 CONF

Diff-MN: Diffusion Parameterized MoE-NCDE for Continuous Time Series Generation with Irregular Observations

ArXiv CS.AI 50%

arXiv:2601.13534v3 Announce Type: replace-cross Abstract: Time series generation (TSG) is widely used across domains, yet most existing methods assume regular sampling and fixed output resolutions. These assumptions are often violated in practice, where observations are...

<https://arxiv.org/abs/2601.13534>

28 CONF

Edit in 2D, Verify in 3D: Reinforcement Learning for Multi-view Consistent Scene Editing

ArXiv CS.AI 50%

arXiv:2603.03143v2 Announce Type: replace-cross Abstract: Leveraging the priors of 2D diffusion models for 3D editing has emerged as a promising paradigm. However, multi-view consistency remains challenging in edited results, and the extreme scarcity of paired 3D-consistent...

<https://arxiv.org/abs/2603.03143>

29 CONF

Test-Time Detoxification without Training or Learning Anything

ArXiv CS.AI 50%

arXiv:2602.02498v2 Announce Type: replace-cross Abstract: Large language models can produce toxic or inappropriate text even for benign inputs, creating risks when deployed at scale. Detoxification is therefore important for safety and user trust, particularly when we want to...

<https://arxiv.org/abs/2602.02498>

30 CONF

ImprovEvolve: Basin-Hopping Meets LLM-Guided Evolutionary Search

ArXiv CS.AI 50%

arXiv:2602.10233v2 Announce Type: replace-cross Abstract: LLM-guided evolutionary computation, most notably AlphaEvolve, has been remarkably successful in discovering novel mathematical constructions by solving challenging optimization problems. The standard approach is to...

<https://arxiv.org/abs/2602.10233>

31 CONF

General and Efficient Steering of Diffusion Models

ArXiv CS.AI 50%

arXiv:2602.11395v2 Announce Type: replace-cross Abstract: Steering diffusion models toward conditions unseen during training typically requires either retraining with conditional inputs or per-step gradient computations, both of which incur substantial computational overhead. We...

<https://arxiv.org/abs/2602.11395>

32 CONF

Choose Your Agent: Tradeoffs in Adopting AI Advisors, Coaches, and Delegates in Multi-Party Negotiation

ArXiv CS.AI 50%

arXiv:2602.12089v3 Announce Type: replace-cross Abstract: As AI usage becomes more prevalent in social contexts, understanding agent-user interaction is critical to designing systems that improve both individual and group outcomes. We present an online behavioral experiment...

<https://arxiv.org/abs/2602.12089>

33 CONF

Mitigating the Safety-utility Trade-off in LLM Alignment via Adaptive Safe Context Learning

ArXiv CS.AI 50%

arXiv:2602.13562v2 Announce Type: replace-cross Abstract: While reasoning models have achieved remarkable success in complex reasoning tasks, their increasing power necessitates stringent safety measures. For safety alignment, the core challenge lies in the inherent trade-off...

<https://arxiv.org/abs/2602.13562>

34 CONF

WoVR: World Models as Reliable Simulators for Post-Training VLA Policies with RL

ArXiv CS.AI 50%

arXiv:2602.13977v2 Announce Type: replace-cross Abstract: Reinforcement learning (RL) promises to unlock capabilities beyond imitation learning for Vision--Language--Action (VLA) models, but its requirement for massive real-world interaction prevents direct deployment on...

<https://arxiv.org/abs/2602.13977>

35 CONF

DyGnROLE: Asymmetric Pretraining for Edge Classification on Dynamic Graphs

ArXiv CS.AI 50%

arXiv:2602.23135v2 Announce Type: replace-cross Abstract: Edge classification on directed dynamic graphs requires modeling interactions between source and destination nodes exhibiting asymmetrical behavioral patterns and temporal dynamics. However, existing dynamic graph...

<https://arxiv.org/abs/2602.23135>

36 CONF

What Capable Agents Must Know: Selection Theorems for Robust Decision-Making under Uncertainty

ArXiv CS.AI 50%

arXiv:2603.02491v3 Announce Type: replace-cross Abstract: As artificial agents become increasingly capable, what internal structure is necessary for an agent to act competently under uncertainty? Classical results show that optimal control can be implemented using belief states...

<https://arxiv.org/abs/2603.02491>

37 CONF

The Hidden Cost of Structured Generation in LLMs: Draft-Conditioned Constrained Decoding

ArXiv CS.AI 50%

arXiv:2603.03305v2 Announce Type: replace-cross Abstract: Large language models (LLMs) are increasingly used to generate executable outputs, JSON objects, and API calls, where a single syntax error can make the output unusable. Constrained decoding enforces validity...

<https://arxiv.org/abs/2603.03305>

38 CONF

Longitudinal Lesion Inpainting in Brain MRI via 3D Region Aware Diffusion

ArXiv CS.AI 50%

arXiv:2603.05693v2 Announce Type: replace-cross Abstract: Accurate longitudinal analysis of brain MRI is often hindered by evolving lesions, which bias automated neuroimaging pipelines. While deep generative models have shown promise in inpainting these lesions, most existing...

<https://arxiv.org/abs/2603.05693>

39 CONF

Proof-of-Guardrail in AI Agents and What (Not) to Trust from It

ArXiv CS.AI 50%

arXiv:2603.05786v2 Announce Type: replace-cross Abstract: As AI agents become widely deployed as online services, users often rely on an agent developer's claim about how safety is enforced, which introduces a threat where safety measures are falsely advertised. To address the...

<https://arxiv.org/abs/2603.05786>

40 CONF

FD^{\$^2}: A Dedicated Framework for Fine-Grained Dataset Distillation

ArXiv CS.AI 50%

arXiv:2603.25144v2 Announce Type: replace-cross Abstract: Dataset distillation (DD) compresses a large training set into a small synthetic set, reducing storage and training cost, and has shown strong results on general benchmarks. Decoupled DD further improves efficiency by...

<https://arxiv.org/abs/2603.25144>

41 CONF

Sustainable Hybrid Document-Routed Retrieval for Financial RAG: Resolving the Robustness-Precision Trade-off

ArXiv CS.AI 50%

arXiv:2603.26815v3 Announce Type: replace-cross Abstract: Retrieval-Augmented Generation (RAG) systems for financial document QA typically follow a chunk-based paradigm: documents are split into fragments, embedded, and retrieved by similarity. In structurally homogeneous...

<https://arxiv.org/abs/2603.26815>

42 CONF

A Knowledge Theory of Capital: The Value of Natural and Artificial Intelligence, Volume 1

ArXiv CS.AI 50%

arXiv:2606.18288v2 Announce Type: replace-cross Abstract: This volume develops a knowledge theory of capital for economies in which productive capacity increasingly resides in software, data, models, routines, expertise, platforms, organizations, commons, and public epistemic...

<https://arxiv.org/abs/2606.18288>

43 CONF

ATMA: Length-Invariant Language Modeling via Polar Attention and Gated-Delta Compression Memory

ArXiv CS.AI 50%

arXiv:2606.25156v2 Announce Type: replace-cross Abstract: Modern large language models based on softmax scaled-dot-product attention are constrained by their training sequence length: as the key-value sequence grows, softmax probability mass can dilute across a wider...

<https://arxiv.org/abs/2606.25156>

44 **CONF****Reclaim Evaluation: A Lossy Memory Is Worse Than an Empty One****ArXiv CS.AI** **50%**

arXiv:2606.25449v2 Announce Type: replace-cross Abstract: A language model's memory can be worse than no memory at all. A memory that keeps a wrong conclusion but drops the work behind it makes the model emit the stale value as a confident answer, where an empty memory would...

<https://arxiv.org/abs/2606.25449>